

IS THERE A PLATONIC TREE?

CALIBRATING LATENT HYPERBOLICITY

IN FOUNDATION MODELS

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ABSTRACT

Hyperbolic representation learning rests on a premise we call latent hyperbolicity: standard models are already tree-like because their class geometry scores a low Gromov δ , a measure of tree-likeness that is zero for a tree. That low value is also what a random cloud of the same dimension and spread scores, so we build an instrument that reads every score as its excess over many such clouds, and a hierarchy test whose ability to detect a hierarchy is measured. We apply it to 12 vision backbones, 6 datasets and 15 text models. On image features, where the premise is read, the calibrated reading is indistinguishable from a random cloud in most model–dataset cells and small elsewhere. Class centroids do carry structure that a random cloud of the same shape does not, in 44 of 72 cells, but so does a star of clusters. In 4 of 12 ImageNet backbones the hierarchy test finds structure, but it is the orientation of clusters relative to the superclass centers, not a hierarchy among those centers: randomizing orientations removes it, whereas a planted hierarchy, synthetic or trained in, survives. Where a planted hierarchy is detected, in 9 of 12 backbones, no hierarchy as strong as the planted one is found; in the other 3 the test cannot tell. MERU, trained in hyperbolic space, shows the same structure as its Euclidean twin in a space that stays nearly flat. In text, the result depends on the model’s recipe and size. Across models, the trees agree on which classes group together well above chance, though less than two readings of the same model, and not on distances; the self-supervised models organize classes by direction rather than by distance, which a comparison by distance misses. Read correctly, foundation models organize classes into clusters that they partly share, show no hierarchy among the superclasses where the test can see one, and their raw tree-likeness is not evidence for hyperbolic geometry.

1 INTRODUCTION

Semantic knowledge is hierarchical, and hyperbolic space is the natural geometric home for hierarchies (Nickel & Kiela, 2017; Sarkar, 2011; Ganea et al., 2018; Chami et al., 2019). A line of work therefore imposes it on representations at training time, from vision to vision–language and language (Khrulkov et al., 2020; Ermolov et al., 2022; Atigh et al., 2022; Bdeir et al., 2024; Desai et al., 2023; Pal et al., 2025; Chen et al., 2022; He et al., 2025; Sinha et al., 2024). These methods cite a premise, which we call *latent hyperbolicity*: standard models already organize concepts in a tree-like metric structure. The evidence is a single number, the Gromov δ of their features, which is zero for a metric tree and grows as a metric departs from one. Measured on CNN and ViT features (Khrulkov et al., 2020; Bdeir et al., 2024), on token embeddings (Yang et al., 2025) and on word vectors (Tifrea et al., 2019), it comes out low. Those readings are taken at the sample level, on individual images or tokens rather than on classes, and they are read against the ideal values of a tree and of a non-hyperbolic space, never against what a structureless cloud of the same dimension and spectrum would score. Is the premise right, and what does it license? Figure 1 frames the question: an observer who sees only a shadow, the raw δ , asks which of three worlds cast it, a structureless cloud, a star of clusters or a tree. All three read alike.

Why is the reading low? Gromov δ assigns zero to trees (Gromov, 1987), but its reference level depends on dimension, spectrum and statistic, so a raw value cannot be called low on its own. High-dimensional clouds concentrate toward equidistance (Beyer et al., 1999; Aggarwal et al., 2001), which is a degenerate star tree: we call this the *dimension confound*, the geometric counterpart of the width confounder of Gröger et al. (2026). Anisotropic spectra lower the effective dimension and



Figure 1: **Plato’s cave: a random cloud, a star and a tree cast the same shadow.** An observer who sees only shadows cannot tell which object cast them, and a raw δ is such a shadow: points with no structure, clusters around a hub, and a nested hierarchy all read as low δ . The instrument reads the objects instead of the shadow: the excess separates the random cloud from the other two, and the hierarchy test separates the star from the tree.

raise the reading: the *spectrum confound*. And the supremum over sampled quadruples grows with the budget and does not converge (Fournier et al., 2015): the *statistic confound*.

We build the instrument that reads δ against the right reference. It calibrates the estimator on reference geometries, expresses every reading as an *excess* over a null cloud matched in dimension and covariance spectrum, ranks that excess against matched replicates, and adds a *hierarchy test* whose false-alarm rate and power are measured on real clouds. With it we ask of geometry what convergence work asks of similarity (Huh et al., 2024; Gröger et al., 2026): what survives calibration, and is it shared? Gröger et al. (2026) calibrate similarity across models; we calibrate geometry within a model, and comparing trees across models needs a calibration of its own.

The answer has three parts. (i) *Where the premise is read*, on image features, the calibrated reading is indistinguishable from a random cloud of the same shape in most model–dataset cells and small elsewhere. The 4 values reported by Khrulkov et al. (2020) are reproduced and, calibrated, 2 of them are indistinguishable from a random cloud. (ii) *Within each model*, class centroids carry structure that a random cloud does not have, in 44 of 72 cells and 30 of 36 on datasets with 47 classes or more, but a star of clusters already produces it. The hierarchy test finds more in 4 of 12 ImageNet backbones: the orientation of each cluster relative to its superclass center, which we call its hub, not a hierarchy among the hubs. Randomizing the orientations removes every verdict, whereas an implanted hierarchy, synthetic or trained in, survives. In the 9 of 12 backbones where an implanted hierarchy is detected, none as strong is found; in the other 3 the test is blind. A backbone trained in hyperbolic space shows the same structure as its Euclidean twin and stays nearly flat. (iii) *Across models*, the trees agree on which classes group together well above chance, though less than two readings of the same model, and not on distances; the self-supervised models organize classes by direction rather than by distance, which a comparison by distance misses.

Our contributions are:

1. **The instrument.** A calibrated reading of δ , compared with random clouds of the same shape, and a hierarchy test whose false-alarm rate and power are measured on real clouds.
2. **The census.** The premise calibrated where it is read, including a published reading reproduced; structure beyond a random cloud in most models; clusters oriented toward their hubs in a few; no hierarchy among the hubs where the test can see one; and a hyperbolic-trained backbone as the control for imposing the geometry.
3. **The map of trees.** Across models, the trees agree on which classes group together but not on distances, and the self-supervised models organize classes by direction, which a comparison by distance misses.
4. **Consequences for practice.** The rule that derives curvature from a raw δ assigns curvature to random clouds; we say what to measure before imposing curvature, and cosine collects most of the structure at no cost.

2 RELATED WORK

Reading δ against a matched reference has precedent in network science, where the δ of real graphs is compared with that of random graphs of comparable size (Narayan & Sanjeev, 2011; Kennedy et al., 2013; Adcock et al., 2013). We bring the idea to embedding clouds, where the reference must match dimension and spectrum. The curvature itself has been studied as a representation tradeoff (Sala et al., 2018) and as a mixed-curvature product to be learned (Gu et al., 2019), which presup-

poses a diagnostic of the kind we calibrate. The analyses of learned representations characterize Euclidean structure and do not test for tree-likeness: representation similarity, intrinsic dimension and neural collapse (Kornblith et al., 2019; Ansuini et al., 2019; Pope et al., 2021; Pappayan et al., 2020). Closest to us is the work on representational convergence: the Platonic Representation Hypothesis (Huh et al., 2024) argues that models converge toward shared representations, and its cross-modal convergence has been stress-tested at scale (Koepke et al., 2026); Gröger et al. (2026) show that permutation-calibrated similarity removes most of that convergence while local neighborhood structure survives, and Park et al. (2025) read concept hierarchy in language models from the orthogonality of concept directions. We are the geometric complement of this program: the same calibration applied within a model instead of across models. It adds a hierarchy test measured on real clouds, a hyperbolic backbone as the control for imposing the geometry, and a calibration of the tree-to-tree comparison itself.

3 METHODOLOGY

3.1 BACKGROUND: GROMOV δ

Tree-likeness is measured on quadruples. Take any four points w, x, y, z of a metric space with distance d . They can be paired in three ways, and each pairing has a sum of two distances:

$$d(w, x) + d(y, z), \quad d(w, y) + d(x, z), \quad d(w, z) + d(x, y). \quad (1)$$

Sort the three sums as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$. In a tree the two largest sums are always equal, because the two pairings that cross the trunk traverse the same path, so the four-point defect $(S_{\max} - S_{\text{mid}})/2$ is zero. The further a metric is from a tree, the larger the defect can become. A metric tree has $\delta = 0$. Euclidean space has no bound at all: enlarging a square enlarges its defect in proportion. Hyperbolic space sits in between, with a bounded defect whose value for the hyperbolic plane is known, $\ln 2$ under this definition (Nica & Špakula, 2016).

Definition 1 (four-point defect). For a quadruple (w, x, y, z) with the pairing sums of Eq. 1 sorted as $S_{\max} \geq S_{\text{mid}} \geq S_{\min}$, the four-point defect is $\delta(w, x, y, z) = \frac{1}{2}(S_{\max} - S_{\text{mid}})$, and the Gromov δ of a metric space is the supremum of the defect over its quadruples (Gromov, 1987).

3.2 ESTIMATION AND NORMALIZATION

Three steps turn δ into a number that can be compared across models. First, a thousand centroids have far more quadruples than can be enumerated, so the defect is computed on half a million random quadruples. Second, the largest defect in that sample is a poor estimate of the supremum: it belongs to a single extreme quadruple and keeps growing as more quadruples are drawn (Fournier et al., 2015). We therefore read the 99.9th percentile of the sampled defects, $\hat{\delta}_{99.9}$, the value that 99.9 per cent of them fall below. It keeps the tail of the defects, as the supremum does, but it is set by hundreds of quadruples rather than by one. It also does not move with the sample size: across quadruple budgets the ImageNet reading drifts by less than half its own spread (Table 3). Third, the defect grows with the size of the point set, so we divide by the diameter, the largest pairwise distance:

$$\delta_{\text{norm}} = \frac{\hat{\delta}_{99.9}}{\text{diam}}. \quad (2)$$

On this scale a tree reads zero, and a hyperbolic region of radius four or more reads below a sphere (Table 4).

Definition 2 (reading of record). The reading of record of a cloud is δ_{norm} of Eq. 2, with $\hat{\delta}_{99.9}$ computed over half a million sampled quadruples per seed and averaged over 10 seeds.

Proposition 1 (range bound and dimension confound). *Let $d_{\min}$ and $d_{\max}$ be the smallest and the largest pairwise distance of a finite metric space. (a) For any quadruple the four-point defect is at most $d_{\max} - d_{\min}$, hence $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$. (b) For an iid Gaussian cloud of n points in dimension d , with n fixed, $d_{\min}/d_{\max} \rightarrow 1$ as d grows (Beyer et al., 1999; Aggarwal et al., 2001), so $\delta_{\text{norm}} \rightarrow 0$ without any structure.*

A raw value cannot be called low on its own. Proposition 1 makes the dimension confound precise; the calibration on reference geometries shows the spectrum and statistic confounds (Table 4).

3.3 SPECTRUM-MATCHED NULL, EXCESS AND GENUINENESS

Definition 3 (Haar null). A null replicate is a cloud with the real shape and no structure. Let $X = U\Sigma V^T$ be the singular value decomposition of the centered cloud; the replicate is $X' = Q\Sigma V^T$ with Q the orthogonal factor of a column-centered Gaussian matrix, orthogonal to the all-ones vector and Haar-distributed on the centered subspace, so X' has exactly the centered singular values of X and structureless coefficients.



Figure 2: **The raw reading cannot be read on its own.** (a) Raw reading and matched null per ImageNet backbone, ordered by dimension; the gap is the excess. (b) Same dataset, same reading, opposite verdict: ViT-B on CIFAR-100 images and DINO-B on CIFAR-100 centroids.

The null is a cloud with the real shape and no structure. Of the two constructions compared, Gaussian coefficients reproduce the spectrum only on average and inflate the excess of low-rank clouds. The centered Haar construction reproduces the centered spectrum exactly, so it is the record. The uncentered and the Gaussian readings are reported beside it (Table 5).

Definition 4 (excess). With K null replicates $X'_1, \dots, X'_K$ of Definition 3, the excess of a cell is

$$e(X) = \delta_{\text{norm}}(X) - \frac{1}{K} \sum_{k=1}^K \delta_{\text{norm}}(X'_k). \quad (3)$$

Size and evidence. The excess is a size; the evidence is the rank of the reading among its replicates and the left-tail p that the rank gives,

$$r = \#\{k : \delta_{\text{norm}}(X'_k) > \delta_{\text{norm}}(X)\}, \quad p = \frac{1 + K - r}{K + 1}, \quad K = 200. \quad (4)$$

Definition 5 (genuine). Order the m left-tail p -values of a census as $p_{(1)} \leq \dots \leq p_{(m)}$. A model-dataset cell is genuine when its p is at or below the Benjamini–Hochberg threshold (Benjamini & Hochberg, 1995) at level five per cent,

$$p \leq p_{(i^*)}, \quad i^* = \max\{i : p_{(i)} \leq 0.05 i/m\}, \quad (5)$$

and the word is used in this sense only.

What a genuine excess means. A negative genuine excess, more than chance would give, certifies structure beyond the second moments; clustering is its most plausible reading, supported by the superclass recovery of Section 5.4. Two cells of the same dataset can have the same reading and opposite verdicts (Figure 2).

3.4 HIERARCHY TEST

Why a second test. A genuine excess can come from clusters alone, so a second test asks whether the clusters are themselves arranged hierarchically. It needs three constructions.

Definition 6 (hub null and excess B). Given a grouping of classes into superclasses, which we call the frame, assigning each centroid x_i to a cluster $k(i)$ with hub $h_{k(i)}$, the cluster mean, write $x_i = h_{k(i)} + o_i$. The hub null keeps every offset o_i and replaces the hubs by a Haar resample with their spectrum, $x'_i = h'_{k(i)} + o_i$, and the excess of Eq. 3 against this null, e_B , measures structure above the clusters of the frame only.

Definition 7 (matched star). The matched star of a cloud keeps the real clusters, each as a Haar sample with the cluster’s own covariance, and places their hubs as Gaussian draws at the real hub radius: the clustering of the real cloud with no hierarchy above it. Its spectrum-matched-hub variant draws the hubs as a Haar resample of the real hubs.

Definition 8 (decoupling control). The decoupling control keeps the hubs of the frame and rotates each cluster’s offsets by an independent Haar rotation. A verdict that survives the decoupling is carried by the arrangement of the hubs; one that does not is carried by the orientation of the clusters relative to their hubs, which we call alignment.

The hierarchy test compares the real cloud with its matched star. With e_B read on the real cloud and on its matched star, the depth statistic and its standardized form are

$$\text{depth} = e_B(X) - e_B(X_{\text{star}}), \quad z = \text{depth} / \text{s.d.}, \quad (6)$$

The spread is taken over star seeds and null replicates, and a cloud is certified against its matched star when $z \leq -2$. The power of the test is the fraction of implanted hierarchies it detects, and its false-alarm rate is the fraction of flat controls that fire. Both are measured on real clouds in Section 5.3.

3.5 COMPARING TREES ACROSS MODELS

Trees are compared under a configuration chosen before the comparison. Two hierarchies are compared by the adjusted Rand index between their cut partitions and by the correlation of their cophenetic distances. They are also compared by their agreement on which pair of a random class triplet merges first. The configuration of metric and linkage is selected per dataset by a criterion fixed without reference to the agreements. It is the highest cophenetic fidelity among configurations whose reference cuts are non-degenerate for every model. Local agreement is read with the permutation calibration of Gröger et al. (2026) on mutual-kNN agreement and centroid CKA.

3.6 PROJECTION AND TASKS

Three readouts are compared on the intact representation. A parameter-free map centers the features, scales the 95th-percentile norm to radius $1/\sqrt{2}$ and preserves directions. Euclidean, cosine and Poincaré readouts are then scored by nearest-centroid classification and 5-way 5-shot episodes on every cell of the task grid.

4 EXPERIMENTAL SETUP

The census has 12 backbones, 6 class sets and 15 text models. The vision backbones are 4 supervised ViTs (Dosovitskiy et al., 2021) trained on ImageNet-21k (Deng et al., 2009), DINO-B (Caron et al., 2021), 4 DINOv2 sizes (Oquab et al., 2024), CLIP-B, CLIP-L (Radford et al., 2021) and SigLIP-B (Zhai et al., 2023). The class sets are ImageNet and CIFAR-100 (Krizhevsky, 2009), which carry a real hierarchy, CIFAR-10 and DTD (Cimpoi et al., 2014), and the flat FMNIST (Xiao et al., 2017) and MNIST (LeCun et al., 1998). The text models read the thousand ImageNet class names: 4 GPT-2 sizes (Radford et al., 2019), 3 Pythia sizes (Biderman et al., 2023), OLMo-1B (Groeneveld et al., 2024) and 7 sentence embedders (Table 6).

Four controls test the alternatives. Two self-supervised ResNets (He et al., 2016) read the vision census. DeiT-B (Touvron et al., 2021) and the augreg ViT-B (Steiner et al., 2022), trained on ImageNet-1k leaf labels, test whether a hierarchical label set explains the alignment. MERU and its Euclidean CLIP twin (Desai et al., 2023) test whether training in hyperbolic space creates depth. The census ViT-B, fine-tuned with and without a hierarchical objective, is the trained control.

The premise is read on images and hierarchy on centroids. The sample-level reading uses about a thousand stratified training images per cell on CIFAR-100 and DTD. Class centroids average a hundred cached training images per class on ImageNet and the full cached split elsewhere. Causal language models give the last-token state of “a photo of a {class}”, extracted one prompt at a time because batching alters GPT-2’s hidden states. Embedders use their recommended pooling (Table 7).

Calibration budget and frames. Each cell is read against 200 Haar replicates, with half a million sampled quadruples per seed and 10 seeds. With 10 classes there are only 210 quadruples, so $\hat{\delta}_{99.9}$ coincides with the supremum there. Benjamini–Hochberg runs over the vision census and, separately, over the text census. The frame, the grouping into superclasses, is the WordNet (Miller, 1995) cut into 30 superclasses on ImageNet and the 20 coarse labels on CIFAR-100. A balanced grouping is the control, with 10 star seeds.

5 RESULTS

5.1 THE PREMISE AT THE SAMPLE LEVEL

The premise does not survive calibration where it is read. On per-image features, where the premise is read, the calibrated reading is not genuine in most cells, that is, no more than chance would give. Where it is genuine, the excess removes at most two fifths of the null reading. Table 8 gives the reading per cell.

A published reading is reproduced and calibrated. Their estimator was run on our extraction; it reproduces the raw δ_{rel} that Khrulkov et al. (2020) report for ResNet-34 on 4 datasets within 0.03. Calibrated, CIFAR-10 and CUB-200 (Wah et al., 2011) are indistinguishable from a random cloud, and CIFAR-100 and MiniImageNet (Vinyals et al., 2016) fall below it. Under their own statistic, the supremum, CIFAR-10 and CUB-200 stay indistinguishable from a random cloud and

dataset	their δ_{rel}	ours, their estimator	excess	$r/200$, largest p	excess (sup.)	$r/200$, p range (sup.)
CIFAR-10	0.26	0.275	-0.0037	173/200, 0.333	-0.0205	167/200, 0.005-0.701
CIFAR-100	0.25	0.261	-0.0087	198/200, 0.030	-0.0353	196/200, 0.005-0.299
CUB-200	0.25	0.270	-0.0043	177/200, 0.249	-0.0218	174/200, 0.005-0.488
MiniImageNet	0.21	0.203	-0.0234	200/200, 0.005	-0.0700	200/200, 0.005-0.005

Table 1: **A published reading reproduced and calibrated.** ResNet-34 setting of Khrulkov et al. (2020): their relative hyperbolicity, ours with their estimator, then the excess and the rank of 200 with the largest left-tail p for the record statistic (99.9th percentile). For their own statistic, the supremum (sup.), against the same replicates: the mean excess, the median rank over 10 trials and the range of p across trials. CIFAR-100 is below the null in the median trial but not in every trial.

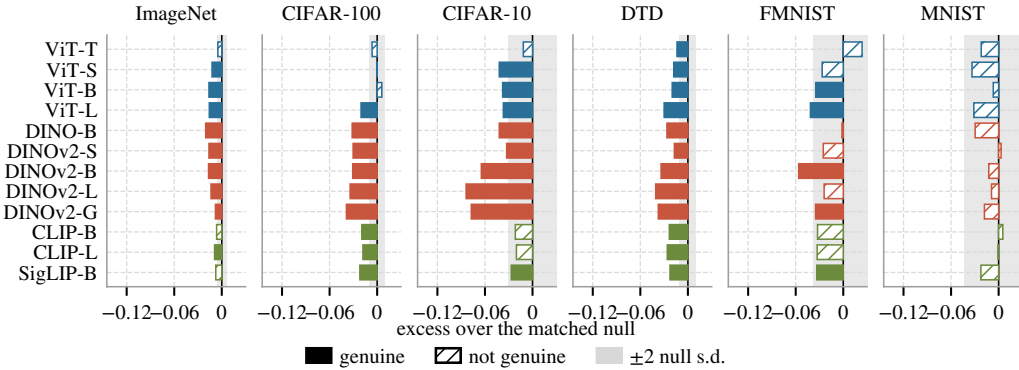


Figure 3: **Most cells show structure beyond the second moments.** Excess of the reading of record over its matched null per dataset and backbone, filled when genuine; band: two null standard deviations.

MiniImageNet stays below it. The verdict for CIFAR-100 changes from trial to trial, as the statistic confound predicts. Table 1 gives the 4 rows.

5.2 STRUCTURE BEYOND THE SECOND MOMENTS AT THE CLASS LEVEL

Most cells show structure beyond the second moments. It is genuine in 44 of 72 cells and 30 of 36 on the 3 datasets with 47 classes or more. On ImageNet and CIFAR-100 the count is 18 of 24. Figure 3 and Table 2 give the census.

The count survives resampling. Folding image resampling and the estimator’s seeds into the null spread, or repeating the census on resampled centroids, keeps most of the count. Table 3 gives the bootstrap.

The structure depends on the class set more than on the model. The smallest backbone is genuine on one dataset only, and the flat datasets have smaller and less consistent excesses; FMNIST is genuine in 5 of 12 backbones. A class-count control shows that the excess shrinks with the number of classes, so only the verdict carries across datasets. Tables 5 and 3 give the details.

A star of clusters already passes the census. A synthetic star with no hierarchy scores an excess as deep as any real cell, so a negative excess says structure beyond the second moments, not depth. Table 4 gives the star’s excess.

Neural collapse is the flat limit, not what the census sees. Neural collapse predicts that supervised class means converge to a simplex frame (Papayan et al., 2020), a star with no hierarchy. The census does not certify that frame: class means cluster around superclass hubs, which the superclass recovery of Section 5.4 shows.

5.3 DEPTH ABOVE THE LABELLED CLUSTERS

The hierarchy test certifies that clusters are oriented toward their hubs in 4 of 12 backbones: ViT-S, ViT-B, ViT-L and DINOv2-L. The test was run on the 12 ImageNet centroid clouds under both matched stars, with the WordNet cut into superclasses as the frame and the superclass centers as hubs. Figure 4a shows the z per backbone and Table 9 gives both stars.

What it certifies is alignment. Once the cluster orientations were randomized, the decoupling control of Definition 8, none of the 4 backbones fires. ViT-L then reads -1.61, within the -1.61 to -1.76 of the flat control: without its orientations the real cloud reads like a flat one. The alignment is relational: 0 of 60 runs fire with random hubs and 0 of 4 with random orientations, so it needs both the real hubs and the real orientations. Removing the radial component of every offset leaves

model	family	$\hat{\delta}_{99.9}$ IN	excess IN	$r/200$ (p)	exc. C100	exc. C10	ρ_{WN}
ViT-T	sup.	0.088	−0.005°	169 (.159)	−0.006°	−0.012°	+0.52
ViT-S	sup.	0.063	−0.012	200 (.005)	−0.000°	−0.042	+0.50
ViT-B	sup.	0.054	−0.016	200 (.005)	+0.006°	−0.038	+0.49
ViT-L	sup.	0.055	−0.016	200 (.005)	−0.020	−0.037	+0.53
DINO-B	SSL	0.055	−0.020	200 (.005)	−0.031	−0.042	+0.36
DINOv2-S	SSL	0.053	−0.016	200 (.005)	−0.030	−0.033	+0.22
DINOv2-B	SSL	0.034	−0.017	200 (.005)	−0.031	−0.065	+0.18
DINOv2-L	SSL	0.030	−0.014	200 (.005)	−0.034	−0.084	+0.18
DINOv2-G	SSL	0.033	−0.008	200 (.005)	−0.039	−0.077	+0.20
CLIP-B	contr.	0.084	−0.007°	172 (.144)	−0.019	−0.022°	+0.57
CLIP-L	contr.	0.076	−0.009	197 (.020)	−0.018	−0.020°	+0.59
SigLIP-B	contr.	0.075	−0.008°	189 (.060)	−0.022	−0.027	+0.58

Table 2: **Structure beyond the second moments is the rule at the class level.** Per backbone: raw ImageNet reading $\hat{\delta}_{99.9}$, excess over the Haar null mean, rank $r/200$ with left-tail p (°: not genuine at BH $p \leq 0.05$ over 72 cells; bold: above the null), the CIFAR-100 and CIFAR-10 excess, and the Spearman correlation ρ_{WN} with WordNet distances.

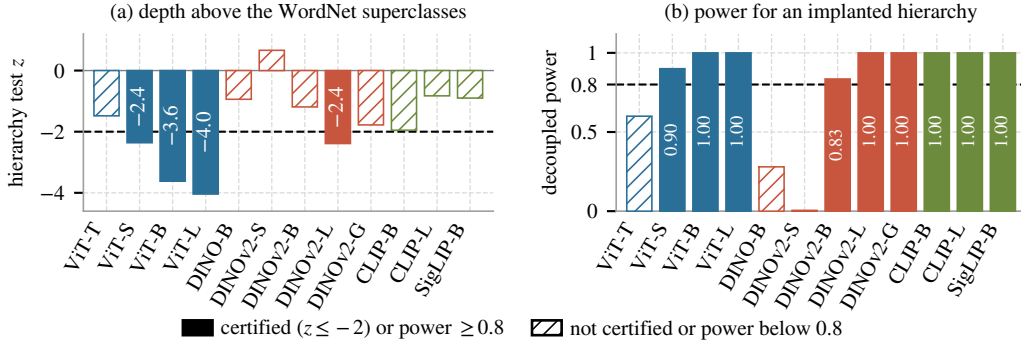


Figure 4: **What the hierarchy test certifies, and where it can see a hierarchy.** (a) Depth z per backbone, filled when certified; line at $z = -2$. (b) Decoupled power for an implanted three-level hierarchy, filled at 0.8 or more; line at 0.8.

it in all 4 certified backbones, z from -2.26 to -3.99 under both stars; feature norms are not its source. Under full L2 normalization, which also moves the hubs, it survives in ViT-B and ViT-L only. The implanted principal-axis alignment reproduces it in the supervised ViTs and CLIP-B but not in DINOv2-L, and its geometric form is not resolved here.

What the test can see. Given an implanted two-level tree at this noise level, the test misses it; given a three-level hierarchy with ViT-L’s spectrum and noise, it detects it in 4 of 5 seeds. That detection survives orientation randomization, whereas none of the 4 real verdicts does. The same hierarchy was then built at each backbone’s own spectrum and ratio. The decoupled control detects it in 9 of 12 backbones (decoupled power 0.83–1.00) and not in the other 3 (power 0.01–0.60). The noise level of a cloud is its within-cluster spread relative to the distance between its hubs. The 3 blind backbones sit at ratios 1.5 to 3.0, inside the 1.3 to 3.9 spanned by the 9 covered backbones. The power therefore follows the backbone, not its ratio or family. ViT-B fine-tuned with a hierarchical cross-entropy keeps firing once decoupled in 20 of 20 runs over 2 seeds, mean $z = -2.51$, on the frame of record. The leaf-only fine-tune fires in 6 of 20, mean $z = -1.37$, and the frozen checkpoint in none. Fine-tuning itself adds some decoupled signal and the hierarchical objective adds more, so the trained control separates the objectives by degree. A flat control with the same spectrum and ratio as the deep one fires in 0 of 5 intact and in 8 of 50 decoupled runs. The decoupled test is therefore biased toward firing, so the 4 real backbones not firing once decoupled is conservative evidence. The synthetic hierarchy at $z = -4.19$ to -4.82 stands well clear of that bias. On the balanced frame the matched flat control fires in 0 of 50 decoupled runs. Tables 10 and 9 give the runs.

No hub hierarchy is found where the test has power. In those 9 backbones, none as strong as the planted one is found. The decoupling separates structure among the hubs of the frame from structure carried by cluster orientations. A hierarchy below the frame expressed in how clusters open would



Figure 5: **The island is the cut; tree topology is shared well above chance, though short of the within-model ceiling; the self-supervised structure is angular.** (a) Mean agreement of each model’s tree with the other 11, naive (hollow) and corrected (filled). (b) Triplet agreement under the corrected comparison against the within-model ceiling (band). (c) Sibling-triplet agreement under cosine and Euclidean distance.

be removed with the orientations and counted as alignment. No hub hierarchy therefore means no hierarchy among the frame’s hubs. Figure 6 and Table 10 give the detection rates and the power.

The certified set depends on the frame. Rerun under a balanced grouping into superclasses, the balanced frame, ViT-T joins the certified set and ViT-S and DINOv2-L leave it; other WordNet cuts change the set again. Only ViT-B and ViT-L are certified under every frame. A flat cloud clustered under the frame of record fires under the balanced frame in 38 of 50 decoupled runs. A real cloud read under a frame that is not its own is therefore expected to fire. The frozen ViT-B firing in 7 of 10 there is consistent with that mismatch and is not evidence of hub structure.

Leaf labels can produce the alignment but do not guarantee it. Read like the census models, the augreg ViT-B trained on ImageNet-1k leaf labels is certified and agrees with WordNet like the ImageNet-21k ViTs. DeiT-B, trained on the same labels with another recipe, is neither. A hierarchical label set therefore does not explain the certified ViTs’ alignment.

MERU’s objective does not create hub structure. MERU embeds images on the Lorentz hyperboloid with an entailment objective, and its Euclidean twin shares backbone, data and recipe (Desai et al., 2023). Read with the census and the hierarchy test, MERU shows the same clustering as its twin, and no hub structure is detected. The test’s power at MERU’s spectrum, however, was not measured. Its embeddings stay in the near-flat regime, 95 per cent of them within 0.29 of the curvature scale. Table 9 gives the readings and the radii.

5.4 WHOSE TREE

The naive map manufactures an island. Figure 5 compares the trees of the 12 models in three ways. Panel (a) compares them as usually done, (b) under the criterion of Section 3.5 against the within-model ceiling, and (c) by whether classes are grouped by direction or by distance. Compared with Euclidean distances and average linkage, the DINOv2 family agrees with the supervised and contrastive block far less than the block agrees with itself. The cause is chaining: DINOv2’s Euclidean dendrograms put almost every class in one cluster, forcing the index to zero.

Controlled, the trees share their topology, not their metric. Across models, triplet agreement reaches 0.77, against 0.90 for two resamples of the same model. Chance gives 0.33. The cophenetic and cut agreements reach about half their within-model ceiling (Table 11).

The self-supervised tree is angular. Sibling triplets were resolved under cosine and under Euclidean distance: DINOv2 resolves them far better under cosine, so its semantics concentrate in the angular component and its Euclidean dendrograms chain. Repeated on cosine geometry, the census agrees with the Euclidean one in most cells, and DINOv2 on ImageNet is genuine, more than chance would give, under both. Figure 5c shows the triplets and Table 5 gives the cosine census.

Agreement with WordNet follows supervision and recipe. The correlation between inter-centroid and WordNet distances is highest for the contrastive VLMs, then the supervised ViTs, and lowest for DINOv2. Supervision deepens the resemblance on average, and DeiT-B shows that the effect is recipe-dependent. Table 12 gives the correlations.

The recovery is not WordNet circularity. On DBpedia Classes (Lehmann et al., 2015), an ontology independent of WordNet, the three text embedders read a genuine excess and recover the true

partition under every configuration. On HierarCaps, a four-level caption hierarchy with no class set (Alper & Averbuch-Elor, 2024), the encoders resolve sibling triplets far above chance. Table 12 gives both.

Text depends on recipe and scale. On the ImageNet class names, 7 of 15 text models are genuine under the record. GPT-2 S is genuine at the margin, GPT-2 M is not, and GPT-2 L and XL are genuine under every reading, as are Pythia at every scale and OLMo-1B. Table 7 gives the census.

5.5 WHAT IS SHARED IS LOCAL

Models share neighborhoods, not metrics. Permutation-calibrated mutual-kNN agreement and centroid CKA survive across all 66 model pairs, and the Poincaré readout improves neither (Gröger et al., 2026) (Table 13).

6 IMPLICATIONS FOR HYPERBOLIC REPRESENTATION LEARNING

The raw reading cannot select a curvature. Khrulkov et al. (2020) set the curvature of the Poincaré ball from the relative hyperbolicity, $c = (0.144/\delta_{\text{rel}})^2$ with $\delta_{\text{rel}} = 2\delta/\text{diam}$. Applied, with the supremum it was calibrated for, to structureless Gaussian clouds, it assigns curvatures from 0.48 to 2.5.

Both readings predict the gain, and the hierarchy verdict predicts none. The raw and calibrated readings predict the gain of a non-Euclidean readout equally well, and the hierarchy verdict predicts none. Calibration tells what a low δ means, not which readout to use.

The calibration certifies structure and does not choose the readout. Cosine collects the self-supervised structure. The Poincaré readout adds +0.9 to +1.3 pp over cosine for the contrastive VLMs on prototype tasks and nothing consistent elsewhere. In few-shot learning a fixed-radius Euclidean encoder does at least as well as hyperbolic prototypes (Moreira et al., 2024) (Table 14).

7 CONCLUSION AND LIMITATIONS

Conclusion. Read correctly, foundation models organize classes into clusters that they partly share, show no hierarchy among the superclasses where the test can see one, and their raw tree-likeness is not evidence for hyperbolic geometry. Curvature should be measured rather than assumed. Cosine collects most of what is there.

Limitations of the instrument. (i) The excess certifies structure beyond second moments, star-like clustering included, so depth needs the separate test. (ii) The spectrum null conditions on the second moments, to which a hierarchy contributes, so a low excess does not exclude hierarchy carried by the spectrum. (iii) The hierarchy test’s power for a three-level hierarchy follows the backbone rather than its within/between ratio or family. Under the decoupled control it is 0.83 or more in 9 of 12 backbones and 0.60 or less in the other 3. Its false alarms on flat clouds are 8 of 50, and the verdict of no hub hierarchy therefore holds in those 9 only. It misses a two-level implant at this noise level, its decoupled form is biased toward firing, and it tests hierarchy among the frame’s hubs only. Its trained control separates the objectives by degree, not absolutely: the leaf-only fine-tune also fires once decoupled, in 6 of 20 runs. A pre-set criterion expecting no intact certification of the leaf and frozen models was not met, because they are aligned. The alignment is not the radial spread of feature norms, which a control removes without changing the verdicts; full L2 normalization keeps it in 2 of the 4 backbones. Its exact form is not resolved here. (iv) The hierarchy test is Euclidean, and for the angular DINOv2 tree it may be conservative. (v) A causal LM’s class representation depends on the extraction protocol as much as on the model. (vi) The analysis rests on many choices, frames, stars and null variants among them, and a single pre-specified analysis is future work. (vii) The reading divides by the diameter, so heavier tails in real clouds would lower δ_{norm} without any clustering. The cosine census mitigates this, and a percentile normalization is future work.

Limitations of scope. (viii) The census is class-centroid geometry on 6 datasets, and the sample-level reading covers 2 datasets, one image budget and one published setting. (ix) The objective-geometry link is correlational, resting chiefly on DINOv2’s scale range and on 2 leaf-label ViTs. (x) Downstream scope is prototype-based metric tasks with modest gains.

REPRODUCIBILITY STATEMENT

Every number in the paper is annotated in the source with the result file that produces it, and all tables are generated by script from those files (Table 15). The full pipeline with fixed seeds will be released in the anonymized repository (TODO (author)). Its `tool/` directory ships the instrument as one script that reproduces any cell of Table 2 from its centroid matrix. Estimator settings are stated in Section 4 and the model panel in Table 6.

ETHICS STATEMENT

This work analyzes publicly released models and standard public datasets, involves no human subjects and no personal data, and proposes an analysis methodology and a metric-selection heuristic. We do not foresee ethical risks beyond those already associated with the underlying models and datasets.

AI USE STATEMENT

In this work, we used generative AI tools for writing assistance; for the retrieval of references; and for research ideation or execution, namely the implementation and execution of experiments under the authors’ direction and LLM-simulated reviews used as methodological feedback. We have not used generative AI tools for other tasks with required disclosure. We have reviewed all AI-assisted work, and we take responsibility for the final content of this work, including text, claims or artifacts produced with the aid of generative AI.

REFERENCES

- Aaron B. Adcock, Blair D. Sullivan, and Michael W. Mahoney. Tree-like structure in large social and information networks. In *IEEE International Conference on Data Mining (ICDM)*, pp. 1–10, 2013. doi: 10.1109/ICDM.2013.77.
- Charu C. Aggarwal, Alexander Hinneburg, and Daniel A. Keim. On the surprising behavior of distance metrics in high dimensional space. In Jan Van den Bussche and Victor Vianu (eds.), *Database Theory - ICDT 2001, 8th International Conference, London, UK, January 4-6, 2001, Proceedings*, volume 1973 of *Lecture Notes in Computer Science*, pp. 420–434. Springer, 2001. doi: 10.1007/3-540-44503-X_27. URL https://doi.org/10.1007/3-540-44503-X_27.
- Morris Alper and Hadar Averbuch-Elor. Emergent visual-semantic hierarchies in image-text representations. In *European Conference on Computer Vision (ECCV)*, 2024. doi: 10.1007/978-3-031-72943-0_13.
- Alessio Ansuini, Alessandro Laio, Jakob H. Macke, and Davide Zoccolan. Intrinsic dimension of data representations in deep neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 6109–6119, 2019. URL <https://proceedings.neurips.cc/paper/2019/hash/cfcce0621b49c983991ead4c3d4d3b6b-Abstract.html>.
- Mina Ghadimi Atigh, Julian Schoep, Erman Acar, Nanne van Noord, and Pascal Mettes. Hyperbolic image segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 4453–4462, June 2022. doi: 10.1109/CVPR52688.2022.00441.
- Ahmad Bdeir, Kristian Schwethelm, and Niels Landwehr. Fully hyperbolic convolutional neural networks for computer vision. In *The Twelfth International Conference on Learning Representations, ICLR 2024, Vienna, Austria, May 7-11, 2024*. OpenReview.net, 2024. URL <https://openreview.net/forum?id=ekzlhN5QNh>.
- Yoav Benjamini and Yosef Hochberg. Controlling the false discovery rate: A practical and powerful approach to multiple testing. *Journal of the Royal Statistical Society: Series B (Methodological)*, 57(1):289–300, 1995. doi: 10.1111/j.2517-6161.1995.tb02031.x.
- Kevin S. Beyer, Jonathan Goldstein, Raghu Ramakrishnan, and Uri Shaft. When is “nearest neighbor” meaningful? In Catriel Beeri and Peter Buneman (eds.), *Database Theory - ICDT ’99, 7th International Conference, Jerusalem, Israel, January 10-12, 1999, Proceedings*, volume 1540 of *Lecture Notes in Computer Science*, pp. 217–235. Springer, 1999. doi: 10.1007/3-540-49257-7_15. URL https://doi.org/10.1007/3-540-49257-7_15.
- Stella Biderman, Hailey Schoelkopf, Quentin Gregory Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. Pythia: A suite for analyzing large language models across training and scaling. In Andreas Krause, Emma Brunskill, Kyunghyun Cho, Barbara Engelhardt, Sivan Sabato, and Jonathan Scarlett (eds.), *International Conference on Machine Learning, ICML 2023, 23-29 July 2023, Honolulu, Hawaii*,

- USA, Proceedings of Machine Learning Research, pp. 2397–2430. PMLR, 2023. URL <https://proceedings.mlr.press/v202/biderman23a.html>.
- Mathilde Caron, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. Emerging properties in self-supervised vision transformers. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 9650–9660, October 2021. doi: 10.1109/ICCV48922.2021.00951.
- Ines Chami, Zhitao Ying, Christopher Ré, and Jure Leskovec. Hyperbolic graph convolutional neural networks. In Hanna M. Wallach, Hugo Larochelle, Alina Beygelzimer, Florence d’Alché-Buc, Emily B. Fox, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 32: Annual Conference on Neural Information Processing Systems 2019, NeurIPS 2019, December 8-14, 2019, Vancouver, BC, Canada*, pp. 4869–4880, 2019.
- Weize Chen, Xu Han, Yankai Lin, Hexu Zhao, Zhiyuan Liu, Peng Li, Maosong Sun, and Jie Zhou. Fully hyperbolic neural networks. In Smaranda Muresan, Preslav Nakov, and Aline Villavicencio (eds.), *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 5672–5686, Dublin, Ireland, May 2022. Association for Computational Linguistics. doi: 10.18653/v1/2022.acl-long.389. URL <https://aclanthology.org/2022.acl-long.389/>.
- Mircea Cimpoi, Subhransu Maji, Iasonas Kokkinos, Sammy Mohamed, and Andrea Vedaldi. Describing textures in the wild. In *2014 IEEE Conference on Computer Vision and Pattern Recognition, CVPR 2014, Columbus, OH, USA, June 23-28, 2014*, pp. 3606–3613. IEEE Computer Society, 2014. doi: 10.1109/CVPR.2014.461. URL <https://doi.org/10.1109/CVPR.2014.461>.
- Jia Deng, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. Imagenet: A large-scale hierarchical image database. In *2009 IEEE Computer Society Conference on Computer Vision and Pattern Recognition (CVPR 2009), 20-25 June 2009, Miami, Florida, USA*, pp. 248–255. IEEE Computer Society, 2009. doi: 10.1109/CVPR.2009.5206848. URL <https://doi.org/10.1109/CVPR.2009.5206848>.
- Karan Desai, Maximilian Nickel, Tanmay Rajpurohit, Justin Johnson, and Ramakrishna Vedantam. Hyperbolic image-text representations. In *Proceedings of the 40th International Conference on Machine Learning, ICML’23*. JMLR.org, 2023. URL <https://proceedings.mlr.press/v202/desai23a.html>.
- Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Szko-reit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. In *International Conference on Learning Representations*, 2021. URL <https://openreview.net/forum?id=YicbFdNTTy>.
- Aleksandr Ermolov, Leyla Mirvakhabova, Valentin Khrulkov, Nicu Sebe, and Ivan V. Oseledets. Hyperbolic vision transformers: Combining improvements in metric learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition, CVPR 2022, New Orleans, LA, USA, June 18-24, 2022*, pp. 7399–7409. IEEE, 2022. doi: 10.1109/CVPR52688.2022.00726. URL <https://doi.org/10.1109/CVPR52688.2022.00726>.
- Hervé Fournier, Anas Ismail, and Antoine Vigneron. Computing the Gromov hyperbolicity of a discrete metric space. *Information Processing Letters*, 115(6-8), 2015. doi: 10.1016/j.ipl.2015.02.002.
- Octavian Ganea, Gary Becigneul, and Thomas Hofmann. Hyperbolic neural networks. In S. Bengio, H. Wallach, H. Larochelle, K. Grauman, N. Cesa-Bianchi, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 31. Curran Associates, Inc., 2018. URL https://proceedings.neurips.cc/paper_files/paper/2018/hash/dbab2adc8f9d078009ee3fa810bea142-Abstract.html.

- Dirk Groeneveld, Iz Beltagy, Evan Pete Walsh, Akshita Bhagia, Rodney Kinney, Oyvind Tafjord, Ananya Harsh Jha, Hamish Ivison, Ian Magnusson, Yizhong Wang, Shane Arora, David Atkinson, Russell Authur, Khyathi Raghavi Chandu, Arman Cohan, Jennifer Dumas, Yanai Elazar, Yuling Gu, Jack Hessel, Tushar Khot, William Merrill, Jacob Morrison, Niklas Muennighoff, Aakanksha Naik, Crystal Nam, Matthew E. Peters, Valentina Pyatkin, Abhilasha Ravichander, Dustin Schwenk, Saurabh Shah, Will Smith, Emma Strubell, Nishant Subramani, Mitchell Wortsman, Pradeep Dasigi, Nathan Lambert, Kyle Richardson, Luke Zettlemoyer, Jesse Dodge, Kyle Lo, Luca Soldaini, Noah A. Smith, and Hannaneh Hajishirzi. Olmo: Accelerating the science of language models. In Lun-Wei Ku, Andre Martins, and Vivek Srikumar (eds.), *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, ACL 2024, Bangkok, Thailand, August 11-16, 2024, pp. 15789–15809. Association for Computational Linguistics, 2024. doi: 10.18653/V1/2024.ACL-LONG.841. URL <https://doi.org/10.18653/v1/2024.acl-long.841>.
- Fabian Gröger, Shuo Wen, and Maria Brbić. Revisiting the Platonic representation hypothesis: An Aristotelian view. In *International Conference on Machine Learning (ICML)*, 2026. doi: 10.48550/ARXIV.2602.14486. URL <https://doi.org/10.48550/arXiv.2602.14486>.
- M. Gromov. Hyperbolic groups. In S. M. Gersten (ed.), *Essays in Group Theory*, pp. 75–263. Springer New York, New York, NY, 1987. ISBN 978-1-4613-9586-7. doi: 10.1007/978-1-4613-9586-7_3. URL https://doi.org/10.1007/978-1-4613-9586-7_3.
- Albert Gu, Frederic Sala, Beliz Gunel, and Christopher Ré. Learning mixed-curvature representations in product spaces. In *International Conference on Learning Representations*, 2019.
- Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *2016 IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pp. 770–778. IEEE, 2016. doi: 10.1109/CVPR.2016.90.
- Neil He, Rishabh Anand, Hiren Madhu, Ali Maatouk, Smita Krishnaswamy, Leandros Tassioulas, Menglin Yang, and Rex Ying. HELM: Hyperbolic large language models via mixture-of-curvature experts. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=RnbJPkakkM>.
- Minyoung Huh, Brian Cheung, Tongzhou Wang, and Phillip Isola. Position: The Platonic representation hypothesis. In Ruslan Salakhutdinov, Zico Kolter, Katherine A. Heller, Adrian Weller, Nuria Oliver, Jonathan Scarlett, and Felix Berkenkamp (eds.), *Forty-first International Conference on Machine Learning, ICML 2024, Vienna, Austria, July 21-27, 2024*, Proceedings of Machine Learning Research, pp. 20617–20642. PMLR / OpenReview.net, 2024. URL <https://proceedings.mlr.press/v235/huh24a.html>.
- W. Sean Kennedy, Onuttom Narayan, and Iraj Saniee. On the hyperbolicity of large-scale networks. *arXiv preprint arXiv:1307.0031*, 2013. URL <https://arxiv.org/abs/1307.0031>.
- Valentin Khrulkov, Leyla Mirvakhabova, Evgeniya Ustinova, Ivan Oseledets, and Victor Lempitsky. Hyperbolic image embeddings. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2020. doi: 10.1109/CVPR42600.2020.00645.
- A. Sophia Koepke, Daniil Zverev, Shiry Ginosar, and Alexei A. Efros. Back into Plato’s cave: Examining cross-modal representational convergence at scale, 2026. URL <https://arxiv.org/abs/2604.18572>. arXiv preprint arXiv:2604.18572.
- Simon Kornblith, Mohammad Norouzi, Honglak Lee, and Geoffrey E. Hinton. Similarity of neural network representations revisited. In Kamalika Chaudhuri and Ruslan Salakhutdinov (eds.), *Proceedings of the 36th International Conference on Machine Learning, ICML 2019, 9-15 June 2019, Long Beach, California, USA*, Proceedings of Machine Learning Research, pp. 3519–3529. PMLR, 2019. URL <http://proceedings.mlr.press/v97/kornblith19a.html>.
- Alex Krizhevsky. Learning multiple layers of features from tiny images. Technical report, University of Toronto, 2009. URL <https://www.cs.toronto.edu/~kriz/learning-features-2009-TR.pdf>.

- Yann LeCun, Léon Bottou, Yoshua Bengio, and Patrick Haffner. Gradient-based learning applied to document recognition. *Proc. IEEE*, 86(11):2278–2324, 1998. doi: 10.1109/5.726791. URL <https://doi.org/10.1109/5.726791>.
- Jens Lehmann, Robert Isele, Max Jakob, Anja Jentzsch, Dimitris Kontokostas, Pablo N. Mendes, Sebastian Hellmann, Mohamed Morsey, Patrick van Kleef, Sören Auer, and Christian Bizer. DBpedia – a large-scale, multilingual knowledge base extracted from Wikipedia. *Semantic Web*, 6(2):167–195, 2015. doi: 10.3233/SW-140134.
- Zehan Li, Xin Zhang, Yanzhao Zhang, Dingkun Long, Pengjun Xie, and Meishan Zhang. Towards general text embeddings with multi-stage contrastive learning. *CoRR*, abs/2308.03281, 2023. doi: 10.48550/ARXIV.2308.03281. URL <https://doi.org/10.48550/arXiv.2308.03281>.
- George A. Miller. WordNet: a lexical database for English. *Communications of the ACM*, 38(11):39–41, 1995. doi: 10.1145/219717.219748.
- Gabriel Moreira, Manuel Marques, João Paulo Costeira, and Alexander Hauptmann. Hyperbolic vs Euclidean embeddings in few-shot learning: Two sides of the same coin. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision (WACV)*, pp. 2082–2090, 2024. doi: 10.1109/WACV57701.2024.00208.
- Onuttom Narayan and Iraj Saniee. Large-scale curvature of networks. *Physical Review E*, 84(6):066108, 2011. doi: 10.1103/PhysRevE.84.066108.
- Bogdan Nica and Ján Špakula. Strong hyperbolicity. *Groups, Geometry, and Dynamics*, 10(3), 2016. doi: 10.4171/GGD/372. arXiv:1408.0250.
- Maximilian Nickel and Douwe Kiela. Poincaré embeddings for learning hierarchical representations. In Isabelle Guyon, Ulrike von Luxburg, Samy Bengio, Hanna M. Wallach, Rob Fergus, S. V. N. Vishwanathan, and Roman Garnett (eds.), *Advances in Neural Information Processing Systems 30: Annual Conference on Neural Information Processing Systems 2017, December 4-9, 2017, Long Beach, CA, USA*, pp. 6338–6347, 2017. URL <https://proceedings.neurips.cc/paper/2017/hash/59dfa2df42d9e3d41f5b02bfc32229dd-Abstract.html>.
- Maxime Oquab, Timothée Darcet, Théo Moutakanni, Huy V. Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, Daniel HAZIZA, Francisco Massa, Alaaeldin El-Nouby, Mido Assran, Nicolas Ballas, Wojciech Galuba, Russell Howes, Po-Yao Huang, Shang-Wen Li, Ishan Misra, Michael Rabbat, Vasu Sharma, Gabriel Synnaeve, Hu Xu, Herve Jegou, Julien Mairal, Patrick Labatut, Armand Joulin, and Piotr Bojanowski. DINOv2: Learning robust visual features without supervision. *Transactions on Machine Learning Research*, 2024. ISSN 2835-8856. URL <https://openreview.net/forum?id=a68SUT6zFt>. Featured Certification.
- Avik Pal, Max van Spengler, Guido Maria D’Amely di Melendugno, Alessandro Flaborea, Fabio Galasso, and Pascal Mettes. Compositional entailment learning for hyperbolic vision-language models. In *The Thirteenth International Conference on Learning Representations*, 2025. URL <https://openreview.net/forum?id=3i13Gev2hV>.
- Vardan Papyan, X. Y. Han, and David L. Donoho. Prevalence of neural collapse during the terminal phase of deep learning training. *Proceedings of the National Academy of Sciences*, 117(40):24652–24663, 2020. doi: 10.1073/pnas.2015509117. URL <https://www.pnas.org/doi/abs/10.1073/pnas.2015509117>.
- Kiho Park, Yo Joong Choe, Yibo Jiang, and Victor Veitch. The geometry of categorical and hierarchical concepts in large language models. In *International Conference on Learning Representations*, 2025. URL <https://arxiv.org/abs/2406.01506>.
- Phillip Pope, Chen Zhu, Ahmed Abdelkader, Micah Goldblum, and Tom Goldstein. The intrinsic dimension of images and its impact on learning. In *9th International Conference on Learning Representations, ICLR 2021, Virtual Event, Austria, May 3-7, 2021*. OpenReview.net, 2021. URL <https://openreview.net/forum?id=XJk19XzGq2J>.

- Alec Radford, Jeffrey Wu, Rewon Child, David Luan, Dario Amodei, and Ilya Sutskever. Language models are unsupervised multitask learners. *OpenAI blog*, 1(8):9, 2019. URL https://cdn.openai.com/better-language-models/language_models_are_unsupervised_multitask_learners.pdf.
- Alec Radford, Jong Wook Kim, Chris Hallacy, Aditya Ramesh, Gabriel Goh, Sandhini Agarwal, Girish Sastry, Amanda Askell, Pamela Mishkin, Jack Clark, Gretchen Krueger, and Ilya Sutskever. Learning transferable visual models from natural language supervision. In Marina Meila and Tong Zhang (eds.), *Proceedings of the 38th International Conference on Machine Learning, ICML 2021, 18-24 July 2021, Virtual Event*, volume 139 of *Proceedings of Machine Learning Research*, pp. 8748–8763. PMLR, 2021. URL <http://proceedings.mlr.press/v139/radford21a.html>.
- Frederic Sala, Christopher De Sa, Albert Gu, and Christopher Ré. Representation tradeoffs for hyperbolic embeddings. In *Proceedings of the 35th International Conference on Machine Learning*, volume 80 of *Proceedings of Machine Learning Research*, pp. 4460–4469. PMLR, 2018. URL <https://proceedings.mlr.press/v80/sala18a.html>.
- Rik Sarkar. Low distortion Delaunay embedding of trees in hyperbolic plane. In Marc J. van Kreveld and Bettina Speckmann (eds.), *Graph Drawing - 19th International Symposium, GD 2011, Eindhoven, The Netherlands, September 21-23, 2011, Revised Selected Papers*, Lecture Notes in Computer Science, pp. 355–366. Springer, 2011. doi: 10.1007/978-3-642-25878-7_34. URL https://doi.org/10.1007/978-3-642-25878-7_34.
- Aditya Sinha, Siqi Zeng, Makoto Yamada, and Han Zhao. Learning structured representations with hyperbolic embeddings. In A. Globerson, L. Mackey, D. Belgrave, A. Fan, U. Paquet, J. Tomczak, and C. Zhang (eds.), *Advances in Neural Information Processing Systems*, volume 37, pp. 91220–91259. Curran Associates, Inc., 2024. doi: 10.52202/079017-2895.
- Andreas Steiner, Alexander Kolesnikov, Xiaohua Zhai, Ross Wightman, Jakob Uszkoreit, and Lucas Beyer. How to train your ViT? data, augmentation, and regularization in vision transformers. *Transactions on Machine Learning Research*, 2022. URL <https://arxiv.org/abs/2106.10270>.
- Alexandru Tifrea, Gary Bécigneul, and Octavian-Eugen Ganea. Poincaré GloVe: Hyperbolic word embeddings. In *International Conference on Learning Representations (ICLR)*, 2019. URL <https://arxiv.org/abs/1810.06546>.
- Hugo Touvron, Matthieu Cord, Matthijs Douze, Francisco Massa, Alexandre Sablayrolles, and Herve Jegou. Training data-efficient image transformers & distillation through attention. In Marina Meila and Tong Zhang (eds.), *Proceedings of the 38th International Conference on Machine Learning*, volume 139 of *Proceedings of Machine Learning Research*, pp. 10347–10357. PMLR, 2021. URL <https://proceedings.mlr.press/v139/touvron21a.html>.
- Oriol Vinyals, Charles Blundell, Timothy Lillicrap, Koray Kavukcuoglu, and Daan Wierstra. Matching networks for one shot learning. In D. Lee, M. Sugiyama, U. Luxburg, I. Guyon, and R. Garnett (eds.), *Advances in Neural Information Processing Systems*, volume 29. Curran Associates, Inc., 2016. URL https://proceedings.neurips.cc/paper_files/paper/2016/hash/90e1357833654983612fb05e3ec9148c-Abstract.html.
- Catherine Wah, Steve Branson, Peter Welinder, Pietro Perona, and Serge Belongie. The Caltech-UCSD birds-200-2011 dataset. Computation & Neural Systems Technical Report CNS-TR-2010-001, California Institute of Technology, 2011. URL <https://authors.library.caltech.edu/records/cvm3y-5hh21>.
- Liang Wang, Nan Yang, Xiaolong Huang, Binxing Jiao, Linjun Yang, Daxin Jiang, Rangan Majumder, and Furu Wei. Text embeddings by weakly-supervised contrastive pre-training. *CoRR*, abs/2212.03533, 2022. doi: 10.48550/ARXIV.2212.03533. URL <https://doi.org/10.48550/arXiv.2212.03533>.
- Han Xiao, Kashif Rasul, and Roland Vollgraf. Fashion-mnist: a novel image dataset for benchmarking machine learning algorithms. *CoRR*, abs/1708.07747, 2017. URL <http://arxiv.org/abs/1708.07747>.

- Shitao Xiao, Zheng Liu, Peitian Zhang, Niklas Muennighoff, Defu Lian, and Jian-Yun Nie. C-pack: Packed resources for general Chinese embeddings. In *Proceedings of the 47th International ACM SIGIR Conference on Research and Development in Information Retrieval, SIGIR '24*, pp. 641–649, New York, NY, USA, 2024. Association for Computing Machinery. ISBN 9798400704314. doi: 10.1145/3626772.3657878. URL <https://doi.org/10.1145/3626772.3657878>.
- Menglin Yang, Ram Samarth B B, Aosong Feng, Bo Xiong, Jiahong Liu, Irwin King, and Rex Ying. Hyperbolic fine-tuning for large language models. In *The Thirty-ninth Annual Conference on Neural Information Processing Systems*, 2025. URL <https://openreview.net/forum?id=TkEdQv0bXB>.
- Xiaohua Zhai, Basil Mustafa, Alexander Kolesnikov, and Lucas Beyer. Sigmoid loss for language image pre-training. In *2023 IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 11941–11952. IEEE, 2023. doi: 10.1109/ICCV51070.2023.01100.

A PROOFS

Proof of Proposition 1. (a) Each pairing sum in Eq. 1 adds two distances of the space, each of which lies in $[d_{\min}, d_{\max}]$. Hence $S_{\max} \leq 2d_{\max}$ and $S_{\min} \geq 2d_{\min}$, so the defect $\frac{1}{2}(S_{\max} - S_{\min})$ is at most $d_{\max} - d_{\min}$ for every quadruple, and so is its supremum δ . Dividing by the diameter $d_{\max}$ gives $\delta_{\text{norm}} \leq 1 - d_{\min}/d_{\max}$; the same bound holds for the 99.9th percentile of the sampled defects, which never exceeds their supremum. (b) Let $x_1, \dots, x_n$ be iid standard Gaussian in $\mathbb{R}^d$ with n fixed. For any pair $i \neq j$ the coordinates of $x_i - x_j$ are iid with variance 2, so $\|x_i - x_j\|^2/(2d) \rightarrow 1$ almost surely as $d \rightarrow \infty$ by the law of large numbers, and the convergence holds jointly over the finitely many pairs. Therefore $d_{\min}/d_{\max} \rightarrow 1$, and by part (a) $\delta_{\text{norm}} \rightarrow 0$. The rate is the distance-concentration rate of Beyer et al. (1999) and Aggarwal et al. (2001): the relative contrast $(d_{\max} - d_{\min})/d_{\min}$ vanishes as $d^{-1/2}$. Part (b) is the dimension confound alone. The statistic confound is separate and elementary: a maximum over a growing sample of quadruples is non-decreasing in the sample size, so the sampled supremum can only rise with the budget. $\square$

B ONE TABLE PER QUESTION

Every table is generated by script from the released per-cell result files, and no value is hand-transcribed; the tables are numbered in the order the main text first cites them, one per question, with their result files indexed in Table 15.

B.1 IS THE READING ROBUST? QUADRUPLE BUDGET, NULL CONSTRUCTION, RESAMPLING AND CLASS COUNT

(a) Quadruple budget under the record (Haar null, p99.9, 200 replicates): the excess of 9 cells at 10^4 to 2×10^6 sampled quadruples per seed; drift = range of the excess over budgets $\geq 10^5$, also divided by the excess's null spread.

model	data	10^4	5×10^4	10^5	5×10^5	10^6	2×10^6	drift ($\geq 10^5$)	drift/s.d.
ViT-L	IN	-0.0161	-0.0161	-0.0160	-0.0157	-0.0159	-0.0159	0.0002	0.07
DINOv2-L	IN	-0.0131	-0.0136	-0.0137	-0.0135	-0.0136	-0.0136	0.0002	0.38
CLIP-B	IN	-0.0063	-0.0060	-0.0059	-0.0065	-0.0064	-0.0064	0.0007	0.13
ViT-L	C100	-0.0208	-0.0208	-0.0206	-0.0205	-0.0205	-0.0205	0.0002	0.03
DINOv2-L	C100	-0.0348	-0.0348	-0.0348	-0.0348	-0.0347	-0.0347	0.0001	0.07
CLIP-B	C100	-0.0186	-0.0195	-0.0188	-0.0195	-0.0194	-0.0194	0.0006	0.10
ViT-L	DTD	-0.0305	-0.0308	-0.0312	-0.0311	-0.0313	-0.0313	0.0002	0.05
DINOv2-L	DTD	-0.0418	-0.0419	-0.0421	-0.0420	-0.0422	-0.0420	0.0002	0.06
CLIP-B	DTD	-0.0243	-0.0235	-0.0242	-0.0240	-0.0243	-0.0242	0.0003	0.05

Table 3: **The excess is budget-stable, the genuine count survives resampling and estimator noise, the excess shrinks with the class count, and training moves it.** (a) Excess against the quadruple budget for 9 cells; drift = range of the record excess over budgets $\geq 10^5$, divided by the excess's spread in the last column; under the record every sign holds at every budget.

(b) Centroid bootstrap under the record (Haar null, p99.9): 30 resamples with replacement of the cached training images per class at the census budget; ImageNet with 20 null replicates per resample, CIFAR-100 and DTD with 200.

model	dataset	images/class	null rep.	excess (reference)	bootstrap mean	bootstrap s.d.	fraction negative
ViT-T	ImageNet	100	20	-0.0055	-0.0051	0.0011	1.00
ViT-S	ImageNet	100	20	-0.0126	-0.0122	0.0007	1.00
ViT-B	ImageNet	100	20	-0.0165	-0.0160	0.0006	1.00
ViT-L	ImageNet	100	20	-0.0160	-0.0160	0.0004	1.00
DINO-B	ImageNet	100	20	-0.0209	-0.0208	0.0003	1.00
DINOv2-S	ImageNet	100	20	-0.0164	-0.0162	0.0004	1.00
DINOv2-B	ImageNet	100	20	-0.0167	-0.0164	0.0002	1.00
DINOv2-L	ImageNet	100	20	-0.0136	-0.0134	0.0001	1.00
DINOv2-G	ImageNet	100	20	-0.0080	-0.0083	0.0004	1.00
CLIP-B	ImageNet	100	20	-0.0068	-0.0069	0.0009	1.00
CLIP-L	ImageNet	100	20	-0.0094	-0.0095	0.0006	1.00
SigLIP-B	ImageNet	100	20	-0.0080	-0.0081	0.0006	1.00
ViT-T	CIFAR-100	500	200	-0.0064	-0.0059	0.0018	1.00
ViT-S	CIFAR-100	500	200	-0.0003	-0.0004	0.0012	0.67
ViT-B	CIFAR-100	500	200	+0.0057	+0.0055	0.0011	0.00
ViT-L	CIFAR-100	500	200	-0.0205	-0.0205	0.0006	1.00
DINO-B	CIFAR-100	500	200	-0.0316	-0.0315	0.0006	1.00
DINOv2-S	CIFAR-100	500	200	-0.0306	-0.0306	0.0004	1.00
DINOv2-B	CIFAR-100	500	200	-0.0313	-0.0313	0.0004	1.00
DINOv2-L	CIFAR-100	500	200	-0.0348	-0.0348	0.0003	1.00
DINOv2-G	CIFAR-100	500	200	-0.0395	-0.0395	0.0002	1.00
CLIP-B	CIFAR-100	500	200	-0.0195	-0.0192	0.0007	1.00
CLIP-L	CIFAR-100	500	200	-0.0181	-0.0183	0.0009	1.00
SigLIP-B	CIFAR-100	500	200	-0.0220	-0.0218	0.0008	1.00
ViT-T	DTD	80	200	-0.0142	-0.0141	0.0016	1.00
ViT-S	DTD	80	200	-0.0186	-0.0189	0.0020	1.00
ViT-B	DTD	80	200	-0.0208	-0.0212	0.0017	1.00
ViT-L	DTD	80	200	-0.0311	-0.0302	0.0017	1.00
DINO-B	DTD	80	200	-0.0272	-0.0267	0.0024	1.00
DINOv2-S	DTD	80	200	-0.0179	-0.0171	0.0025	1.00
DINOv2-B	DTD	80	200	-0.0350	-0.0345	0.0016	1.00
DINOv2-L	DTD	80	200	-0.0420	-0.0417	0.0016	1.00
DINOv2-G	DTD	80	200	-0.0386	-0.0387	0.0026	1.00
CLIP-B	DTD	80	200	-0.0240	-0.0242	0.0021	1.00
CLIP-L	DTD	80	200	-0.0266	-0.0270	0.0022	1.00
SigLIP-B	DTD	80	200	-0.0233	-0.0234	0.0014	1.00

Table 3: (continued) (b) Bootstrap of the excess over resampled images per class on ImageNet, CIFAR-100 and DTD: the bootstrap s.d. is at most 0.0026 and every cell keeps the sign of its record excess in at least 0.67 of its resamples.

(c) Joint sensitivity of the genuine count: z_{joint} against null, bootstrap and estimator noise, and (in parentheses) the number of the 30 centroid resamples in which the cell is genuine.

model	IN	C100	C10	DTD	FMNIST	MNIST
ViT-T	-1.1 (0) [°]	-0.7 (0) [°]	-0.6 (0) [°]	-1.9 (12) [†]	+1.3 (0) [°]	-0.9 (0) [°]
ViT-S	-3.6 (30)	-0.0 (0) [°]	-2.4 (29)	-3.0 (30)	-1.5 (30) ^{°‡}	-1.3 (0) [°]
ViT-B	-5.2 (30)	+0.8 (0) [°]	-2.1 (1) [†]	-3.9 (30)	-1.9 (30) [†]	-0.3 (0) [°]
ViT-L	-5.0 (30)	-4.0 (30)	-2.4 (30)	-6.4 (30)	-2.2 (30)	-1.4 (0) [°]
DINO-B	-8.9 (30)	-7.6 (30)	-2.4 (30)	-4.4 (30)	-0.1 (0) [°]	-1.4 (0) [°]
DINOv2-S	-15.9 (30)	-9.7 (30)	-2.2 (30)	-3.1 (30)	-1.3 (0) [°]	+0.2 (0) [°]
DINOv2-B	-31.7 (30)	-14.7 (30)	-4.9 (30)	-8.0 (30)	-2.6 (30)	-0.5 (0) [°]
DINOv2-L	-30.7 (30)	-21.6 (30)	-6.0 (30)	-10.6 (30)	-1.3 (30) ^{°‡}	-0.4 (0) [°]
DINOv2-G	-11.5 (30)	-28.2 (30)	-6.5 (30)	-8.2 (30)	-1.8 (23) [†]	-0.7 (0) [°]
CLIP-B	-1.3 (0) [°]	-3.1 (30)	-1.3 (0) [°]	-3.5 (30)	-1.4 (0) [°]	+0.3 (0) [°]
CLIP-L	-2.1 (11) [†]	-3.5 (30)	-1.3 (0) [°]	-4.2 (30)	-1.5 (0) [°]	-0.0 (0) [°]
SigLIP-B	-1.7 (0) [°]	-4.3 (30)	-1.7 (29) [†]	-4.0 (30)	-1.8 (30) [†]	-1.3 (0) [°]

Table 3: (continued) (c) Per cell: $z_{\text{joint}} = \text{excess} / \sqrt{\sigma_{\text{null}}^2 + \sigma_{\text{boot}}^2 + \sigma_{\text{est}}^2}$ and the number of resamples in which the cell is genuine under BH over the 72 cells; [°]: not genuine in the record; [†]: record-genuine but failing one of the two criteria ($z_{\text{joint}} \leq -2$, genuine in ≥ 27 of 30); [‡]: not record-genuine but passing one. Counts: record 44/72 (18/24); $z_{\text{joint}} \leq -2$: 39/72 (18/24); genuine in ≥ 27 of 30 resamples: 42/72 (17/24).

(d) Class count: ImageNet subsets of C classes, random (spanning the hierarchy) or WordNet-coherent (siblings, effectively flat), means over the subset seeds. The excess shrinks with the number of classes for coherent and random subsets alike, so magnitudes are not comparable across class counts; the verdict is what carries across datasets.

model	C	random subsets (span the hierarchy)				WordNet-coherent (siblings)			
		$\hat{\delta}_{99.9}$	excess	below	NC adv	$\hat{\delta}_{99.9}$	excess	below	NC adv
DINOv2-L	10	0.030	-0.134	5/5	+1.04	0.068	-0.098	5/5	+0.08
DINOv2-L	20	0.034	-0.091	5/5	+1.32	0.071	-0.065	5/5	+0.12
DINOv2-L	50	0.032	-0.046	5/5	+1.58	0.050	-0.042	5/5	+0.53
DINOv2-L	100	0.031	-0.032	5/5	+1.46	0.044	-0.031	5/5	+0.77
DINOv2-L	200	0.031	-0.022	5/5	+1.45	0.041	-0.023	5/5	+0.59
DINOv2-L	500	0.030	-0.016	2/2	+1.01	0.031	-0.018	2/2	+0.82
DINOv2-L	1000	0.030	-0.014	1/1	+0.57	—	—	—	—

Table 3: (continued) (d) Class-count control under the record (means over subset seeds): the excess shrinks with the number of classes for coherent and random subsets alike, so magnitudes are not comparable across class counts and the verdict is what carries across datasets; “below” = subset seeds whose real value lies below the null at uncorrected $p \leq 0.05$; NC adv = prototype-classifier advantage of the Poincaré readout in pp.

(e) Training moves the raw reading within an architecture: one row per intervention. Statistic: the raw supremum δ_{norm} , the largest four-point defect over sampled quadruples divided by the diameter, on 512 sampled points; fine-tuning row: 1000 CIFAR-100 test images before and after a non-hierarchical (colour) fine-tuning of DINOv2-S, 5×10^4 quadruples; depth row: DINOv2-B hidden states per layer.

intervention	model	δ before	δ after	change
non-hierarchical fine-tuning	DINOv2-S	0.095	0.161	+69%
across depth (layer 1 $\rightarrow$ 12)	DINOv2-B	0.147	0.058	−61%

Table 3: (continued) (e) Raw δ within architecture: fine-tuning on a task without class hierarchy raises it and it falls across transformer depth.

B.2 IS A LOW RAW READING EVIDENCE? CALIBRATION ON REFERENCE GEOMETRIES AND SYNTHETIC CLUSTERS

(a) Reference geometries.

space	absolute δ	δ_{norm}
balanced binary tree (depth 10)	0.000	0.000
$\mathbb{H}^2$ ($K = -1$), region radius $R=2/4/8/16$	0.65/0.69/0.69/0.69	0.162/0.087/0.043/0.022
uniform S^{99} (chord / geodesic)	0.25/0.37	0.143/0.179
iid Gaussian ($n=1000$), $d=192, \dots, 1536$	—	0.104/0.081/0.074/0.061/0.055/0.046

Table 4: **A low raw δ is not evidence of hierarchy, and the spectrum excess certifies structure beyond the second moments, not depth.** (a) Calibration on reference geometries: the estimator recovers the tree zero and the four-point constant of $\mathbb{H}^2$, spheres stay high, and iid Gaussians drift low as d grows, the dimension confound; the ordering trees $<$ hyperbolic $<$ spherical holds for hyperbolic regions of radius $R \geq 4$. Gaussian row: the sampled supremum from the released control file, the statistic the curvature rule uses; tree/ $\mathbb{H}^2$ /sphere rows re-verified synthetically at every freeze. (b) Excess of synthetic clouds (means over seeds; within/between noise ratio in parentheses) under the spectrum null A, with Gaussian coefficients and with Haar-rotated coefficients (exact sample spectrum), and under the hub-randomizing null B, which keeps every cluster and resamples the hub configuration. A pure star sits far below null A under either construction, so null A certifies structure beyond the second moments, not depth; null B also reports the star as hierarchical, since a near-regular simplex of hubs already minimizes δ , which is why the hierarchy test of Table 9 is read against a matched star rather than against null B. Under the Haar construction the two-level hierarchy sits 2–3 $\times$ below the star; under the plain Gaussian null the mid-radius star scores -0.104 and the 6 $\times$ 5 hierarchy -0.14 , a fraction of either’s excess.

(b) Synthetic clusters: what each null certifies.

synthetic cloud ($n=1000, d=768$)	$\hat{\delta}$	spectrum null A		hub-randomizing null B	
		Gaussian	Haar	Gaussian	Haar
30-cluster star, tight (0.1)	0.055	-0.115	-0.115	-0.126	-0.058
30-cluster star, mid (0.3)	0.059	-0.104	-0.103	-0.114	-0.049
30-cluster star, loose (0.6)	0.061	-0.084	-0.082	-0.090	-0.033
2-level hierarchy 6×5 (0.3)	0.044	-0.143	-0.143	-0.113	-0.112

Table 4: (continued)

B.3 IS THE STRUCTURE BEYOND THE SECOND MOMENTS GENUINE? THE VISION CENSUS UNDER EVERY CONSTRUCTION

What each null preserves. The spectrum-matched null, Gaussian or Haar, preserves n , d , the mean and the covariance spectrum and destroys all higher-order structure; the PC-permutation null preserves each principal component’s marginal and destroys the dependence across components; the label shuffle keeps the point cloud itself and destroys the embedding-label correspondence; and the hub-randomizing null keeps each cluster’s geometry and hub assignment and destroys the hub arrangement, which is why it serves the hierarchy test only when read against a matched star. “Genuine” in the text always means beyond the first of these. In plain terms, the spectrum null is a cloud with the same shape as the real one and no structure inside it; the hub null of Section 5 keeps every cluster of the real cloud and only rearranges where the clusters sit.

(a) The census of record per cell: Haar spectrum-matched null with Q orthogonal to the all-ones vector, 99.9th-percentile statistic, 200 replicates.

model	IN			C100			C10			DTD			FMNIST			MNIST		
	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p	exc.	r	p
ViT-T	-0.005°	169	.159	-0.006°	151	.249	-0.012°	148	.264	-0.014	195	.030	+0.024°	19	.905	-0.022°	168	.164
ViT-S	-0.012	200	.005	-0.000°	112	.443	-0.042	199	.010	-0.018	199	.010	-0.027°	192	.045	-0.034°	183	.090
ViT-B	-0.016	200	.005	+0.006°	42	.791	-0.038	196	.025	-0.020	200	.005	-0.035	198	.015	-0.007°	122	.393
ViT-L	-0.016	200	.005	-0.020	200	.005	-0.037	200	.005	-0.030	200	.005	-0.041	200	.005	-0.031°	187	.070
DINO-B	-0.020	200	.005	-0.031	200	.005	-0.042	199	.010	-0.026	200	.005	-0.002°	104	.483	-0.030°	182	.095
DINOv2-S	-0.016	200	.005	-0.030	200	.005	-0.033	200	.005	-0.017	200	.005	-0.025°	184	.085	+0.003°	79	.607
DINOv2-B	-0.017	200	.005	-0.031	200	.005	-0.065	200	.005	-0.034	200	.005	-0.056	199	.010	-0.012°	142	.294
DINOv2-L	-0.014	200	.005	-0.034	200	.005	-0.084	200	.005	-0.041	200	.005	-0.024°	186	.075	-0.009°	135	.328
DINOv2-G	-0.008	200	.005	-0.039	200	.005	-0.077	200	.005	-0.037	200	.005	-0.035	197	.020	-0.018°	163	.189
CLIP-B	-0.007°	172	.144	-0.019	200	.005	-0.022°	184	.085	-0.023	200	.005	-0.032°	185	.080	+0.005°	81	.597
CLIP-L	-0.009	197	.020	-0.018	200	.005	-0.020°	180	.104	-0.026	200	.005	-0.033°	186	.075	-0.001°	97	.517
SigLIP-B	-0.008°	189	.060	-0.022	200	.005	-0.027	197	.020	-0.023	200	.005	-0.033	195	.030	-0.022°	176	.124

Table 5: **Structure beyond the second moments is the rule at the class level under every construction of the null and statistic, and the verdict does not depend on the metric.** (a) Per cell: excess of $\hat{\delta}_{99.9}$ over the null mean, r = replicates above the real value, $p = (1 + \#\{\text{null} \leq \text{real}\})/201$; genuine = Benjamini-Hochberg-corrected $p \leq 0.05$ over the 72 cells (°: not genuine). Genuine cells: 44/72 overall, 18/24 on ImageNet+CIFAR-100; sign agreement with the original 5-replicate Gaussian $\times$ supremum census on the 72 comparable cells: 65/72. (b) Five symbols per cell in the order centered Haar $\times$ p99.9 (the record, $Q \perp 1$), uncentered Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum, • = genuine under that construction; genuine counts 44/72, 49/72, 46/72, 52/72, 52/72. The percentage is the record excess divided by the mean null reading (ImageNet -33% to -5%; all cells -75% to +18%); the last column is the ImageNet excess of the supremum statistic under the same Haar null with its rank. The centered Haar construction reproduces the centered spectrum exactly, the uncentered one differs by a term of order $1/\sqrt{n}$ that moves the verdict only on the ten-class datasets; the Gaussian one does not at small n and inflates excesses for low-rank clouds (Table 3). The supremum is decided by a few extreme quadruples and disagrees with the 99.9th percentile in both directions on DINOv2-S/B/G ImageNet.

(b) Verdict under the five null $\times$ statistic constructions and the excess as a fraction of the null reading.

model	IN	C100	C10	DTD	FMNIST	MNIST	sup. IN exc. (r)
ViT-T	○○●● -5%	○○●● -5%	○○○○ -9%	●●●● -11%	○○○○ +18%	○○○○ -14%	-0.020 (199)
ViT-S	●●●● -16%	○○●● -0%	●●●● -33%	●●●● -16%	●●●● -21%	○○○○ -21%	-0.020 (200)
ViT-B	●●●● -23%	○○○○ +5%	●●●● -30%	●●●● -18%	●●●● -26%	○○○○ -5%	-0.022 (200)
ViT-L	●●●● -22%	●●●● -18%	●●●● -31%	●●●● -28%	●●●● -31%	○○●● -20%	-0.027 (200)
DINO-B	●●●● -27%	●●●● -28%	●●●● -32%	●●●● -22%	○○○○ -1%	○○○○ -21%	+0.000 (96)
DINOv2-S	●●●● -23%	●●●● -31%	●●●● -28%	●●●● -14%	○○●● -19%	○○○○ +2%	+0.017 (0)
DINOv2-B	●●●● -33%	●●●● -38%	●●●● -59%	●●●● -31%	●●●● -40%	○○○○ -8%	+0.018 (0)
DINOv2-L	●●●● -31%	●●●● -48%	●●●● -75%	●●●● -37%	●●●● -19%	○○○○ -6%	+0.006 (0)
DINOv2-G	●●●● -20%	●●●● -56%	●●●● -75%	●●●● -34%	●●●● -26%	○○○○ -11%	+0.020 (0)
CLIP-B	○○●● -7%	●●●● -16%	○○●● -17%	●●●● -19%	○○○○ -19%	○○○○ +4%	-0.023 (199)
CLIP-L	●●●● -11%	●●●● -16%	●●●● -17%	●●●● -21%	○○○○ -22%	○○○○ -0%	-0.016 (198)
SigLIP-B	○○●● -9%	●●●● -18%	●●●● -23%	●●●● -19%	●●●● -23%	○○●● -16%	-0.015 (194)

Table 5: (continued)

(c) The same census on cosine geometry (L2-normalized centroids, geodesic distances, Haar null on the normalized cloud): robustness, not the record.

model	IN		C100		C10		DTD		FMNIST		MNIST	
	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r	exc.	r
ViT-T	-0.012	200	-0.014 ^o	191	-0.043	198	-0.017	199	+0.007 ^o	73	-0.026 ^o	167
ViT-S	-0.017	200	-0.017	200	-0.073	200	-0.021	200	-0.049	200	-0.036 ^o	183
ViT-B	-0.022	200	-0.006 ^o	170	-0.066	200	-0.021	200	-0.041	199	-0.010 ^o	130
ViT-L	-0.021	200	-0.024	200	-0.080	200	-0.024	200	-0.066	200	-0.043	196
DINO-B	-0.021	200	-0.035	200	-0.070	200	-0.024	200	-0.006 ^o	116	-0.038 ^o	193
DINOv2-S	-0.014	200	-0.033	200	-0.084	200	-0.018	200	-0.054	200	-0.004 ^o	113
DINOv2-B	-0.013	200	-0.032	200	-0.140	200	-0.021	200	-0.081	200	-0.019 ^o	147
DINOv2-L	-0.010	200	-0.029	200	-0.151	200	-0.024	200	-0.051	200	-0.020 ^o	145
DINOv2-G	-0.006	200	-0.029	200	-0.169	200	-0.017	200	-0.064	200	-0.023 ^o	157
CLIP-B	-0.017	200	-0.023	200	-0.039	197	-0.031	200	-0.038 ^o	185	+0.002 ^o	93
CLIP-L	-0.017	200	-0.025	200	-0.045	199	-0.031	200	-0.038 ^o	192	-0.008 ^o	124
SigLIP-B	-0.013	200	-0.025	200	-0.053	200	-0.027	200	-0.052	199	-0.044	197

Table 5: (continued) (c) Cosine reading, BH over 72 cells: sign-negative 70/72, genuine 56/72 and 22/24 on ImageNet+CIFAR-100, verdict agreement with the Euclidean record 60/72 cells. Text under the same cosine protocol (padding-free embeddings): genuine 8/15: gpt2-m, gpt2-l, gpt2-xl, pythia-410m, pythia-1b, pythia-2b8, olmo-1b, gte-qwen2. The angular tree: sibling-triplet agreement on CIFAR-100 for DINOv2-L is 0.91 under cosine against 0.71 under Euclidean distance. (d) Two self-supervised ResNet-50 backbones read the same clustered structure as the ViTs.

(d) Backbones outside the ViT panel on the ImageNet centroid store (spectrum null, 20 replicates).

model	d	$\hat{\delta}$	excess (z)	reps above
Barlow Twins (ResNet-50)	2048	0.075	-0.039 (-4.8)	20/20
BYOL (ResNet-50)	2048	0.073	-0.020 (-2.4)	20/20

Table 5: (continued)

B.4 THE MODEL PANEL AND EXTRACTION

type	model	params
Supervised ViT (vision)	ViT-T (i21k)	5M
Supervised ViT (vision)	ViT-S (i21k)	22M
Supervised ViT (vision)	ViT-B (i21k)	86M
Supervised ViT (vision)	ViT-L (i21k)	307M
SSL (vision)	DINO-B	86M
SSL (vision)	DINOv2-S	22M
SSL (vision)	DINOv2-B	86M
SSL (vision)	DINOv2-L	307M
SSL (vision)	DINOv2-G	1.1B
Contrastive (vision)	CLIP-B	86M
Contrastive (vision)	CLIP-L	307M
Contrastive (vision)	SigLIP-B	86M
Causal LM	GPT-2 S	117M
Causal LM	GPT-2 M	345M
Causal LM	GPT-2 L	774M
Causal LM	GPT-2 XL	1.5B
Causal LM	Pythia-410M	410M
Causal LM	Pythia-1B	1.0B
Causal LM	Pythia-2.8B	2.8B
Causal LM	OLMo-1B	1.0B
Text embedder	BGE-base (Xiao et al., 2024)	110M
Text embedder	BGE-large	335M
Text embedder	GTE-base (Li et al., 2023)	110M
Text embedder	GTE-large	335M
Text embedder	GTE-Qwen2-1.5B	1.5B
Text embedder	E5-base (Wang et al., 2022)	110M
Text embedder	E5-large	335M
Controls (vision)	DeiT-B/16 (IN-1k, no distillation)	86M
Controls (vision)	ViT-B/16 augreg (IN-1k)	86M
Controls (vision)	MERU ViT-S/B/L and CLIP twins	22M/86M/307M
Controls (vision)	Barlow Twins, BYOL (ResNet-50)	24M

Table 6: The model panel. 12 vision backbones (all in the census; the 10 with cached per-dataset features, all but DINO-B and SigLIP-B, form the task grid), 8 causal LMs, 7 text embedders, and the vision models used only as controls. Class centroids average 100 cached training images per class on ImageNet and the full cached split, 80–6,000 images per class, on the other datasets. Feature extraction: vision backbones use the encoder’s default pooled embedding (timm, num_classes=0, CLS or global pool per architecture); causal LMs use the last-token hidden state, extracted one prompt at a time (padding-free: left-padded batching alters GPT-2’s hidden states, Table 7); text embedders use their recommended pooling (CLS for BGE, mean for GTE/E5, last token for GTE-Qwen2); CLIP/SigLIP text towers use their text encoders.

B.5 IS TEXT THE SAME? THE TEXT CENSUS UNDER THE RECORD

Extraction, anisotropy and templates. Extraction is part of the measurement: left-padded batching moves GPT-2’s $\hat{\delta}$ by up to 0.018, of the order of the fragile excesses; the padding-free protocol is the record. Genuineness is not a proxy for anisotropy: GTE-base has the highest effective rank and is not genuine, and the only LLM-backed embedder tested (GTE-Qwen2-1.5B) is not genuine under the record. Causal-LM values are template-sensitive but the scale ordering is template-robust (Table 7).

(a) The census of record on the 1000 ImageNet class prompts: padding-free extraction, Haar null, 99.9th-percentile statistic, 200 replicates.

model	type	d	$\hat{\delta}_{99.9}$	excess	record exc./null	$r/200$	p	2×2	cosine exc.	r
GPT-2 S	causal LM	768	0.077	-0.013	-15%	197	.020	●●●○	+0.000°	101
GPT-2 M	causal LM	1024	0.052	-0.005°	-9%	170	.154	○○○○	-0.019	200
GPT-2 L	causal LM	1280	0.053	-0.023	-30%	200	.005	●●●●	-0.016	200
GPT-2 XL	causal LM	1600	0.048	-0.027	-36%	200	.005	●●●●	-0.018	200
Pythia-410M	causal LM	1024	0.067	-0.018	-21%	200	.005	●●●●	-0.024	200
Pythia-1B	causal LM	2048	0.066	-0.015	-18%	199	.010	●●●●	-0.026	200
Pythia-2.8B	causal LM	2560	0.066	-0.017	-20%	200	.005	●●●●	-0.023	200
OLMo-1B	causal LM	2048	0.051	-0.024	-32%	200	.005	●●●●	-0.032	200
BGE-base	embedder	768	0.078	+0.003°	+4%	26	.871	○○○○	-0.002°	151
BGE-large	embedder	1024	0.078	+0.002°	+3%	49	.756	○○○○	-0.003°	170
GTE-base	embedder	768	0.076	+0.004°	+6%	11	.945	○○○○	+0.001°	79
GTE-large	embedder	1024	0.075	+0.003°	+3%	36	.821	○○○○	-0.001°	131
GTE-Qwen2-1.5B	embedder	1536	0.078	+0.002°	+3%	58	.711	○○○○	-0.006	199
E5-base	embedder	768	0.074	+0.004°	+6%	15	.925	○○○○	+0.000°	97
E5-large	embedder	1024	0.071	+0.002°	+3%	44	.781	○○○○	-0.001°	119

Table 7: **In text, structure beyond the second moments depends on recipe and scale: genuine at every Pythia scale and at the larger GPT-2 sizes, at the margin for GPT-2 S, absent for GPT-2 M and for the sentence embedders on this probe.** (a) r = replicates above the real value, p the left-tail add-one p -value; genuine = BH-corrected $p \leq 0.05$ over the 15 models (°: not genuine); exc./null: excess as a fraction of the null reading; the 2×2 column gives the verdicts under Haar $\times$ p99.9, Haar $\times$ supremum, Gaussian $\times$ p99.9, Gaussian $\times$ supremum (● genuine); cosine: the same census on cosine geometry. Genuine 7/15; GPT-2 S is genuine at $p = .020$ with excess -0.013 and GPT-2 M is not ($p = .154$, -0.005), while L and XL are genuine under every construction. GTE-base has the highest effective rank and is not genuine, so genuineness is not a proxy for anisotropy; the only LLM-backed embedder (GTE-Qwen2-1.5B) is not genuine. (b) Causal-LM values are template-sensitive (range of raw $\hat{\delta}$ up to 0.065 for causal LMs against ≤ 0.012 for embedders; “gloss only” contains no class name) but the scale ordering is template-robust: with only 3 null replicates per cell the starred S and M cells are borderline, as the 200-replicate record in (a) shows, whereas the L/XL cells are not in doubt. (c) Precision is irrelevant (fp16 equals fp32 to four decimals); for GPT-2, left-padded batching changes the hidden states (mean cosine to the padding-free extraction 0.98) and moves $\hat{\delta}$ by up to 0.018, while OLMo handles left padding correctly and is invariant, which is why padding-free extraction is the record.

(b) Prompt templates: raw $\hat{\delta}$ per template and, for the four GPT-2 sizes, the template-conditioned excess over the spectrum null (3 replicates; *: $|z| \geq 2$) and its fraction of the null.

template	raw $\hat{\delta}$				GPT-2 excess (padding-free)				GPT-2 excess / null			
	BGE-b	E5-b	GPT-2 S	GPT-2 M	S	M	L	XL	S	M	L	XL
photo	0.124	0.114	0.131	0.091	-0.016*	-0.019*	-0.043*	-0.050*	-11%	-19%	-34%	-41%
name_only	0.118	0.119	0.145	0.122	-0.006	-0.007	-0.048*	-0.051*	-6%	-5%	-39%	-43%
image	0.125	0.117	0.110	0.110	-0.018*	-0.028*	-0.050*	-0.049*	-13%	-26%	-41%	-41%
closeup	0.126	0.120	0.086	0.100	+0.007	-0.018*	-0.044*	-0.049*	+5%	-18%	-36%	-40%
this_is	0.125	0.115	0.080	0.134	-0.032*	-0.022*	-0.042*	-0.046*	-19%	-22%	-34%	-38%
wild	0.115	0.116	0.089	0.104	-0.023*	-0.067*	-0.014	-0.010*	-16%	-43%	-11%	-8%
def_pair	0.119	0.108	0.105	0.081	-0.002	-0.019*	-0.036*	-0.035*	-1%	-21%	-28%	-29%
gloss_only	0.115	0.115	0.124	0.107	-0.020*	-0.030*	-0.033*	-0.036*	-14%	-31%	-27%	-29%
concept	0.119	0.120	0.112	0.112	-0.002	-0.003	-0.043*	-0.041*	-1%	-3%	-38%	-38%
discussion	0.121	0.116	0.105	0.098	-0.010	-0.013*	-0.041*	-0.040*	-7%	-17%	-35%	-35%
mean over templates					-0.012	-0.023	-0.040	-0.041				
templates at $ z \geq 2$					5 of 10	8 of 10	9 of 10	10 of 10				

Table 7: (continued)

(c) Extraction: the same 1000 prompts, only batch size and precision vary.

batch size	GPT-2 M		OLMo-1B	
	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$	fp32 $\hat{\delta}$	fp16 $\hat{\delta}$
1 (no padding)	0.0816	0.0817	0.0817	0.0817
16	0.0996	0.0997	0.0817	0.0816
32	0.0905	0.0905	0.0817	0.0817

Table 7: (continued)

B.6 DOES THE PREMISE SURVIVE AT THE SAMPLE LEVEL? THE CENSUS ON PER-IMAGE FEATURES

(a) The census cells at the sample level.

model	CIFAR-100 (10 images/class, $n=1000$)					DTD (22 images/class, $n=1034$)				
	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	exc./null	$r/200$ (p)	sup. $\hat{\delta}$	$\hat{\delta}_{99.9}$	excess	exc./null	$r/200$ (p)
ViT-T	0.130	0.093	-0.0016°	-2%	134 (0.333)	0.124	0.087	-0.0004°	-0%	111 (0.448)
ViT-S	0.121	0.086	$+0.0051^\circ$	+6%	21 (0.896)	0.103	0.070	-0.0062°	-8%	197 (0.020)
ViT-B	0.121	0.082	$+0.0031^\circ$	+4%	50 (0.751)	0.092	0.060	-0.0080°	-12%	200 (0.005)
ViT-L	0.100	0.067	-0.0041°	-6%	185 (0.080)	0.092	0.056	-0.0073°	-12%	200 (0.005)
DINO-B	0.104	0.070	-0.0042°	-6%	194 (0.035)	0.110	0.075	$+0.0008^\circ$	+1%	80 (0.602)
DINOv2-S	0.100	0.065	-0.0049°	-7%	200 (0.005)	0.145	0.090	$+0.0058^\circ$	+7%	0 (1.000)
DINOv2-B	0.089	0.050	-0.0090°	-15%	200 (0.005)	0.139	0.073	-0.0019°	-3%	162 (0.194)
DINOv2-L	0.087	0.043	-0.0131°	-24%	200 (0.005)	0.131	0.064	-0.0060°	-9%	200 (0.005)
DINOv2-G	0.102	0.036	-0.0232°	-39%	200 (0.005)	0.136	0.065	-0.0079°	-11%	200 (0.005)
CLIP-B	0.117	0.080	-0.0060°	-7%	184 (0.085)	0.125	0.089	$+0.0021^\circ$	+2%	71 (0.647)
CLIP-L	0.132	0.084	$+0.0037^\circ$	+5%	35 (0.826)	0.120	0.082	-0.0008°	-1%	124 (0.383)
SigLIP-B	0.113	0.078	-0.0074°	-9%	199 (0.010)	0.114	0.079	-0.0044°	-5%	184 (0.085)

Table 8: **The sample-level reading, the object of prior latent-hyperbolicity claims, sits within null noise in most cells.** The instrument on ≈ 1000 stratified training images per cell (per-image features, not centroids): sup. $\hat{\delta}$ is the raw supremum δ_{norm} the literature reports; $\hat{\delta}_{99.9}$, excess, exc./null, $r/200$ and p follow Table 2 (Haar spectrum-matched null on the sample cloud, 99.9th-percentile statistic, 200 replicates); $^\circ$: not genuine under Benjamini–Hochberg over the 24 cells. The raw supremum lies in the band 0.087–0.145; the excess is not genuine in 14 of 24 cells and genuine in 10 (DINOv2-S, DINOv2-B, DINOv2-L, DINOv2-G, SigLIP-B on CIFAR-100; ViT-S, ViT-B, ViT-L, DINOv2-L, DINOv2-G on DTD), where it reaches at most -0.023 , that is -39% to -7% of the null reading.

(b) A published reading reproduced and calibrated: the setting of Khrulkov et al. (2020), Table 1, ResNet-34 row.

dataset	their δ_{rel}	our δ_{rel}	excess	rank r	largest p
CIFAR-10	0.26	0.275 ± 0.017	-0.0037 ± 0.0017	173/200	0.333
CIFAR-100	0.25	0.261 ± 0.013	-0.0087 ± 0.0025	198/200	0.030
CUB-200	0.25	0.270 ± 0.012	-0.0043 ± 0.0020	177/200	0.249
MiniImageNet	0.21	0.203 ± 0.013	-0.0234 ± 0.0018	200/200	0.005

(b) Penultimate features of an ImageNet-pretrained ResNet-34 (torchvision, no substitution) on class-balanced batches of 1500 points; their estimator (exact δ on the batch, $\delta_{\text{rel}} = 2\delta/\text{diam}$; mean $\pm$ s.d. over 10 batches) next to their published value, and on the same clouds the record instrument: excess over the centered Haar null (200 replicates, 99.9th-percentile statistic), the mean rank of the reading among the replicates and the largest left-tail p over the 10 batches. Every reproduced raw value is within 0.03 of the published one.

Table 8: (continued)

B.7 IS THE STRUCTURE DEEPER THAN A STAR? THE HIERARCHY TEST

Real-data readings by regime. ImageNet ($n=1000$, WordNet $K=30$) lies in the regime where false alarms are controlled: the real cloud is more hierarchical than its anisotropic matched star beyond the noise ($z \leq -2$) in 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L), never less, under Gaussian and under Haar-resampled star hubs alike (Table 9, Figure 4a). CIFAR-100 ($n=100$, $K=20$) lies in the regime where false alarms are not controlled, where pure stars are declared more hierarchical in 17% of runs; its readings (3 of 12 with $z \leq -2$, none with $z \geq +2$) are reported without certification (Table 9). Validity domain of the test as run: $n \approx 1000$; at $n=100$ the false-alarm rate is 0% with $K=6$ (17 points per cluster) and 17% with $K \geq 12$, so the test is not validated below about ten points per cluster. The bar was fixed before the sweep; the restriction to the regime where false alarms are controlled was adopted after it.

A trained positive control. Table 9(d) gives the fine-tuned ViT-B/16 controls, cross-entropy alone and with a hierarchical term, and the decoupled comparison that separates them.

(a) The 12 backbones under the stars with Gaussian hubs: depth = excess B of the real centroids minus that of a matched star (negative = more hierarchical above the frame), with z against the combined spread; the WordNet cut at $K=10, 30$ and 60 superclasses and the balanced frame.

model	C100, $K=20$		ImageNet, anisotropic star				bal. frame
	iso. z	aniso. depth (z)	iso. $K=30$ z	$K=10$ z	$K=30$ depth (z)	$K=60$ z	real z
ViT-T	+0.02	-0.025 (-1.43)	-1.48	-1.70	-0.018 (-1.48)	-2.53	-2.08
ViT-S	+2.34	-0.008 (-0.51)	-1.95	-1.72	-0.028 (-2.36)	-1.78	-1.72
ViT-B	+4.47	+0.003 (+0.23)	-2.87	-2.57	-0.033 (-3.62)	-2.23	-2.82
ViT-L	+4.20	-0.017 (-1.49)	-3.17	-2.73	-0.033 (-4.04)	-4.69	-5.57
DINO-B	+0.86	-0.038 (-4.24)	-0.21	-0.62	-0.009 (-0.94)	-1.11	-0.98
DINOv2-S	+0.55	-0.020 (-1.61)	+0.94	+0.26	+0.008 (+0.66)	+0.61	+0.12
DINOv2-B	+5.40	+0.010 (+0.94)	-0.93	-0.46	-0.016 (-1.19)	-0.75	-0.32
DINOv2-L	+6.77	+0.013 (+1.20)	-2.27	-1.20	-0.028 (-2.39)	-1.25	-1.70
DINOv2-G	+3.76	-0.007 (-0.68)	-1.57	-1.00	-0.029 (-1.78)	-0.48	-0.93
CLIP-B	+1.75	-0.024 (-2.00)	-0.72	-1.55	-0.016 (-1.95)	-2.10	-1.29
CLIP-L	+5.68	-0.013 (-2.34)	+0.46	-1.26	-0.008 (-0.83)	-1.40	-0.68
SigLIP-B	+2.77	-0.021 (-2.17)	+0.05	-0.76	-0.011 (-0.90)	-1.66	-1.12

Table 9: **The hierarchy test certifies in 4 of 12 ImageNet backbones under both matched stars that clusters are oriented toward their hubs, not that the hubs of the frame form a hierarchy, and no hub hierarchy is found in 9 of 12 backbones where the decoupled control detects an implanted one, and the test is blind in the other 3: none fires once cluster orientations are randomized, whereas a three-level hierarchy implanted at the same noise level is detected and survives that randomization (Table 10); imposing the geometry does not create the structure, and leaf-label supervision can produce it or not.** (a) Isotropic star: Gaussian clouds with each superclass’s RMS spread (the earlier, withdrawn construction); anisotropic star: within each superclass a Haar sample with the cloud’s own covariance, hubs Gaussian at the real hub radius; Haar-resampled hubs: the same clusters with the hubs drawn as a Haar resample of the real hubs. Hub-randomizing Haar null, 10 star seeds; frames: CIFAR-100 coarse labels, ImageNet WordNet cut. The test controls false alarms at $n=1000$ only (Table 10); CIFAR-100 readings ($n=100$) are reported without certification. Certified on ImageNet ($n=1000$, the regime where false alarms are controlled): 4 of 12 backbones (ViT-S, ViT-B, ViT-L, DINOv2-L) with z from -2.36 to -4.04 ; the withdrawn isotropic star reached $z=+6.77$ on CIFAR-100. With the hubs of the matched star drawn as a Haar resample of the real hubs (exact hub spectrum, 10 star seeds) the certified set is ViT-S, ViT-B, ViT-L, DINOv2-L, with z from -2.04 to -3.73 ; r_{star} counts the star seeds whose excess B lies at or below the real one, a rank whose resolution is one part in eleven and which does not separate the certified backbones from the others. Balanced frame (complete-linkage WordNet cut with the smallest variance of cluster sizes, chosen before any depth run): certified ViT-T, ViT-B, ViT-L. K sweep: CIFAR-100 $K=5$: iso 0/12 at $z \leq -2$, 7/12 at $z \geq +2$; aniso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=10$: iso 0/12 at $z \leq -2$, 9/12 at $z \geq +2$; aniso 1/12 at $z \leq -2$, 0/12 at $z \geq +2$. CIFAR-100 $K=20$: iso 0/12 at $z \leq -2$, 8/12 at $z \geq +2$; aniso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=10$: iso 0/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 2/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=30$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. ImageNet $K=60$: iso 3/12 at $z \leq -2$, 0/12 at $z \geq +2$; aniso 4/12 at $z \leq -2$, 0/12 at $z \geq +2$. (b) Supervision on leaf labels can produce the depth (the augreg recipe is certified under both stars) and need not (DeiT-B is not detected): the hierarchical label set of IN-21k is not what makes the census ViTs deep; their WordNet alignment is in Table 12.

(a') The same backbones under the star whose hubs are a Haar resample of the real hubs (ImageNet, $K=30$); last column: the decoupling control, the real hubs kept and each cluster's offsets rotated by an independent Haar rotation (mean $z \pm$ s.d. over 10 seeds; seeds certified at $z \leq -2$).

model	star B	depth (z)	r_{star}	decoupled z (cert.)
ViT-T	+0.0011	-0.0184 (-1.45)	0/10	-1.69 $\pm$ 0.28 (1/10)
ViT-S	+0.0010	-0.0257 (-2.04)	0/10	-1.30 $\pm$ 0.19 (0/10)
ViT-B	+0.0012	-0.0347 (-3.01)	0/10	-1.40 $\pm$ 0.25 (0/10)
ViT-L	+0.0000	-0.0337 (-3.73)	0/10	-1.61 $\pm$ 0.19 (0/10)
DINO-B	+0.0002	-0.0092 (-1.02)	0/10	+1.75 $\pm$ 0.16 (0/10)
DINOv2-S	+0.0007	+0.0083 (+0.71)	10/10	+1.38 $\pm$ 0.14 (0/10)
DINOv2-B	+0.0018	-0.0155 (-1.18)	0/10	+0.41 $\pm$ 0.11 (0/10)
DINOv2-L	-0.0007	-0.0267 (-2.21)	0/10	-0.20 $\pm$ 0.14 (0/10)
DINOv2-G	-0.0021	-0.0263 (-1.50)	0/10	-0.04 $\pm$ 0.07 (0/10)
CLIP-B	+0.0012	-0.0171 (-1.48)	1/10	-1.25 $\pm$ 0.09 (0/10)
CLIP-L	+0.0018	-0.0090 (-0.74)	2/10	-0.88 $\pm$ 0.22 (0/10)
SigLIP-B	+0.0024	-0.0121 (-0.78)	1/10	-0.84 $\pm$ 0.18 (0/10)

Table 9: (continued)

(b) Two ViT-B/16 supervised on ImageNet-1k leaf labels only, through the census protocol, against the IN-21k census backbone.

model	IN excess (r)	C100 excess (r)	depth, Gaussian hubs (z)	depth, Haar hubs (z)
DeiT-B, IN-1k	-0.0073 (200)	-0.0257 (200)	-0.0213 (-1.26)	-0.0197 (-1.21)
ViT-B, augreg, IN-1k	-0.0070 (199)	-0.0117 (187)	-0.0249 (-3.03)	-0.0290 (-2.65)
ViT-B, augreg, IN-21k (census)	-0.0162 (200)	+0.0057 (43)	-0.0335 (-3.62)	-0.0347 (-3.01)

Table 9: (continued)

(c) A hyperbolic backbone and its Euclidean twin under the census of record and the hierarchy test.

		CIFAR-100 ($K=20$, $n=100$: false alarms uncontrolled)			ImageNet ($K=30$, $n=1000$: false alarms controlled)		
model	input	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z	exc. (r)	$\hat{\delta}_{99.9}$ nat./Eucl.	depth z
MERU ViT-S	images	−0.020 (199)	0.117/0.117	−1.52	−0.008° (186)	0.088/0.088	−1.54
CLIP ViT-S	images	−0.028 (200)	0.106/0.107	−3.95	−0.006° (171)	0.110/0.088	−1.85
MERU ViT-B	images	−0.020 (200)	0.114/0.114	−2.16	−0.010 (196)	0.087/0.087	−0.86
CLIP ViT-B	images	−0.030 (200)	0.105/0.103	−3.28	−0.009° (187)	0.107/0.085	−0.99
MERU ViT-L	images	−0.012° (195)	0.118/0.118	−0.29	−0.007° (184)	0.088/0.088	−0.52
CLIP ViT-L	images	−0.016 (199)	0.123/0.112	−1.68	−0.010 (197)	0.104/0.084	−1.21
MERU ViT-S	prompts	−0.022 (200)	0.086/0.086	−0.67	−0.003° (155)	0.085/0.086	+0.02
CLIP ViT-S	prompts	−0.023 (200)	0.085/0.082	−0.78	+0.006° (15)	0.093/0.087	+0.82
MERU ViT-B	prompts	−0.012 (200)	0.096/0.096	−0.70	−0.001° (132)	0.089/0.090	+0.76
CLIP ViT-B	prompts	−0.017 (200)	0.094/0.088	+0.31	+0.008° (2)	0.100/0.092	+0.18
MERU ViT-L	prompts	−0.006° (190)	0.102/0.102	+0.58	+0.001° (75)	0.093/0.093	+0.52
CLIP ViT-L	prompts	−0.012 (200)	0.098/0.092	+0.48	+0.006° (5)	0.101/0.092	+0.39

Table 9: (continued) (c) MERU (Desai et al., 2023) embeds images and text on the Lorentz hyperboloid with an entailment objective; its released twin is a CLIP baseline with the same ViT, RedCaps data and recipe minus the hyperbolic lift and entailment loss. Both pass through the instrument unchanged: class centroids of the projected embeddings (MERU: space-like hyperboloid coordinates; CLIP: unit vectors), Euclidean distances; exc. (r): excess over the Haar spectrum-matched null with the number of 200 replicates above the real value (°: not genuine under BH over the 24 cells); nat./Eucl.: the same statistic in the model’s own metric (Lorentz distance between tangent-space-mean centroids for MERU, angular for CLIP) against the Euclidean one; depth z : the anisotropic matched-star test of panel (a) (10 star seeds), validated at $n=1000$ only. On ImageNet images the excess runs from −0.007 to −0.010 for MERU and from −0.006 to −0.010 for its twin across ViT-S/B/L; MERU’s Lorentz-metric $\hat{\delta}_{99.9}$ differs from its Euclidean value by at most 0.0002; depth z runs from −1.54 to −0.52 for MERU and from −1.85 to −0.99 for CLIP, so neither is more hierarchical than a matched star and MERU is never ahead of its twin. Its embeddings sit in the near-flat regime of the hyperboloid: learned curvature $c=0.10$, spatial norm times $\sqrt{c}$ at the median 0.258–0.279 (95th percentile 0.263–0.287), and Lorentz-to-Euclidean pairwise distance ratio 0.9972–0.9991 at the median (0.9988–0.9997 on the class centroids), so the control tests the training objective, not a curved geometry. MERU’s hierarchy is a generic→specific (text⊃image) partial order, not a class taxonomy. (d) Five epochs, AdamW, lr 10^{-5} encoder and 10^{-3} heads, batch 256, seeds 0 and 1; centroids of the census subset; decoupling over 10 seeds (fraction certified in parentheses). The hierarchical model is certified on both frames and keeps firing once decoupled in both seeds; the leaf-CE model and the frozen checkpoint are certified intact, the frozen checkpoint does not fire once decoupled on the frame of record and the leaf-CE model fires there in 0 of 10 decoupled seeds for seed 0 and 6 of 10 for seed 1, and the frozen checkpoint fires in 7 of 10 decoupled seeds on the balanced frame. The pre-set criterion (leaf-CE and frozen not certified intact) is not met, because ViT-B is certified by alignment; the discriminating comparison is the decoupled one.

(d) A trained positive control: the census ViT-B/16 fine-tuned on the full ImageNet-1k training set with leaf cross-entropy alone and with an added hierarchical cross-entropy at the WordNet 30/6/2 cuts (weights 1/2/4), 2 seeds, identical batches within a seed; census excess under the centered Haar null, hierarchy test and decoupling control on both frames.

ViT-B/16	excess (r)	z , WordNet-30	dec. z (cert.)	z , balanced	dec. z (cert.)
frozen (census)	−0.0162 (200)	−3.62	−1.40 (0.0)	−2.82	−2.10 (0.7)
leaf cross-entropy, seed 0	−0.0243 (200)	−2.45	−0.72 (0.0)	−2.09	−1.38 (0.0)
leaf + hierarchical, seed 0	−0.0293 (200)	−2.97	−2.38 (1.0)	−3.57	−3.51 (1.0)
leaf cross-entropy, seed 1	−0.0242 (200)	−3.18	−2.02 (0.6)	−2.94	−3.02 (1.0)
leaf + hierarchical, seed 1	−0.0295 (200)	−4.22	−2.63 (1.0)	−5.82	−3.44 (1.0)

Table 9: (continued)

(e) Radial control: the hierarchy test on the same clouds after L2-normalizing every centroid and after removing the radial component of every offset (its projection onto the hub direction), under both matched stars, with the decoupling control on each transformed cloud.

model	L2-normalized			radial component removed		
	z aniso	z Haar-hub	decoupled z (cert.)	z aniso	z Haar-hub	decoupled z (cert.)
ViT-T	+0.27	-0.17	-0.78 (0.0)	-2.06	-2.07	-1.74 (0.3)
ViT-S	-0.76	-0.93	-1.16 (0.0)	-2.26	-2.08	-1.25 (0.0)
ViT-B	-2.45	-2.52	-0.57 (0.0)	-3.60	-3.57	-1.40 (0.0)
ViT-L	-2.00	-2.23	-1.41 (0.1)	-3.99	-3.47	-1.47 (0.0)
DINO-B	+0.82	+0.95	+2.00 (0.0)	-1.61	-1.89	+1.91 (0.0)
DINOv2-S	+1.27	+1.38	+1.28 (0.0)	+0.61	+0.58	+1.43 (0.0)
DINOv2-B	-0.07	-0.01	+1.28 (0.0)	-1.18	-1.19	+0.46 (0.0)
DINOv2-L	-0.46	-0.50	+0.89 (0.0)	-2.59	-2.28	-0.01 (0.0)
DINOv2-G	-0.78	-0.76	+0.88 (0.0)	-1.75	-1.66	+0.11 (0.0)
CLIP-B	-1.42	-0.93	-1.38 (0.0)	-1.53	-1.06	-1.19 (0.0)
CLIP-L	-1.57	-1.12	-0.82 (0.0)	-1.05	-0.89	-0.92 (0.0)
SigLIP-B	-1.13	-0.88	-0.51 (0.0)	-1.12	-1.07	-0.89 (0.0)

Table 9: (continued)

B.8 HOW MUCH DEPTH CAN THE TEST SEE? POWER ON SYNTHETIC AND IMPLANTED HIERARCHIES

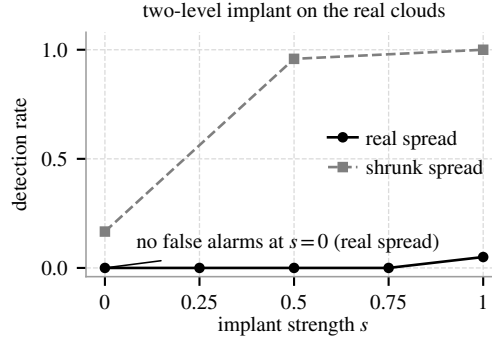


Figure 6: **A two-level implant on the real ImageNet clouds is missed at the real within-cluster spread and detected once the spread is shrunk.** Detection rate at $z \leq -2$ against implant strength s with the real within-cluster spread (solid) and shrunk (dashed): at full strength 0.05 against 1.00, and 0.00 at $s = 0$ with the real spread.

(a) Synthetic hierarchies at $d=768$: two- and three-level hierarchies (pooled) and pure stars, within/between noise ratios 0.1/0.3/0.6, isotropic and anisotropic clusters pooled, 5 seeds.

frame	n	K	power @ ratio 0.1	0.3	0.6	star $z \leq -2$	star $z \geq +2$
top-level, isotropic star	100	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	12	0.00	0.00	0.00	0.00	0.13
top-level, isotropic star	100	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	100	30	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	6	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	12	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	20	0.00	0.00	0.00	0.00	0.17
top-level, isotropic star	1000	30	0.00	0.00	0.00	0.00	0.17
leaf clusters, anisotropic star	100	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	100	12	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	100	20	1.00	1.00	1.00	0.17	0.00
leaf clusters, anisotropic star	1000	6	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	12	1.00	1.00	1.00	0.00	0.00
leaf clusters, anisotropic star	1000	20	1.00	1.00	0.90	0.00	0.00
leaf clusters, anisotropic star	1000	30	1.00	1.00	0.70	0.00	0.00

Table 10: **The hierarchy test has full power on synthetic hierarchies at the leaf frame and none at the top-level frame used on real backbones.** (a) Power = fraction of hierarchy runs with $z \leq -2$; star columns = false-alarm rates in each direction. Top-level frame: the test receives the K super-cluster labels, as the earlier isotropic test did on the real backbones; its hub null keeps every frame cluster intact and only rearranges the K hubs, so hierarchy below the frame is invisible by construction. Leaf frame: the test receives the finest cluster labels with the anisotropic matched star (configurations with more leaves than points are infeasible and omitted). Pre-set bar: power ≥ 0.8 for two- and three-level hierarchies at noise ratios ≤ 0.3 for both n , and false alarms $\leq 5\%$ in each direction. Leaf frame: met at $n=1000$ (power 1.00 at ratios ≤ 0.3 and 0.90 at 0.6; false alarms 0% and 0%), failed at $n=100$, where $K \geq 12$ (under ten points per cluster) gives 17% false alarms at $z \leq -2$. Top-level frame: power 0.00, false alarms 16% pooled. (b) Six super-hubs of five, hub = s -super-hub + $(1 - s + 0.4s)$ -own Gaussian draw, hub cloud rescaled to the real RMS radius; the hierarchy test of Table 9 runs unchanged, 5 implant seeds per s . hits: implant seeds with $z \leq -2$ (of 5); s^* : first s with ≥ 4 of 5; within/between: RMS of the within-cluster offsets over the point-weighted RMS of the hub displacements (1.3 to 3.9 on the real clouds, against 0.1–0.6 in the synthetic sweep); tight: the same implant after shrinking the offsets to within/between = 0.6 (2 seeds). Power (fraction of runs with $z \leq -2$) by s : $s=0$: 0.00, $s=0.25$: 0.00, $s=0.5$: 0.00, $s=0.75$: 0.00, $s=1$: 0.05; false alarms at $s=0$: 0 of 60. Under the Haar-hub star the same implants give 0 of 60 false alarms at $s=0$ and 1 of 60 detections at $s=1$ (power 0.02). Balanced frame: the $K=30$ agglomerative cut of the WordNet distance matrix (complete linkage) with the smallest variance of cluster sizes, chosen and logged before any depth run; sizes 1–154 classes against 1–316 for the frame of record. (c) Implanted alignment is detected in 36 of 60 runs at full strength (60%) and in 1 of 60 at zero; the pre-set bar of 0.8 power at full strength is not met, and the power is backbone-dependent: every seed for the supervised ViTs and CLIP-B, none for DINOv2-S/B/L.

(b) Implanted two-level trees on the real ImageNet centroids: the hub arrangement replaced at strength s , every within-cluster offset kept; hits = implant seeds with $z \leq -2$ at $s=0/0.5/1$, $\bar{z}_1$ = mean z at $s=1$, w/b = within/between spread.

model	WordNet-30 frame of record, Gaussian-hub star						Haar-hub star		balanced frame (pre-specified)		
	real z	w/b	hits	s^*	$\bar{z}_1$	tight z at $s=0/0.5/1$	hits $s=0/1$	$\bar{z}_1$	real z	hits	s^*
ViT-T	-1.48	1.5	0/0/0	-	+0.12	-0.60/-1.85/-9.22	0/0	+0.04	-2.08	0/0/1	-
ViT-S	-2.36	1.9	0/0/0	-	-1.22	-0.23/-3.45/-10.05	0/0	-0.70	-1.72	0/0/0	-
ViT-B	-3.62	2.0	0/0/0	-	+0.37	-1.03/-3.92/-11.45	0/0	+0.41	-2.82	0/0/0	-
ViT-L	-4.04	1.9	0/0/0	-	-1.37	-1.61/-4.37/-12.68	0/0	-0.98	-5.57	0/0/0	-
DINO-B	-0.94	2.1	0/0/0	-	-0.31	-1.61/-5.14/-12.09	0/0	-0.29	-0.98	0/0/0	-
DINOv2-S	+0.66	3.0	0/0/0	-	-0.79	-0.19/-3.06/-11.48	0/0	-0.79	+0.12	0/0/0	-
DINOv2-B	-1.19	3.7	0/0/0	-	+0.41	-0.01/-3.07/-11.17	0/0	+0.55	-0.32	0/0/0	-
DINOv2-L	-2.39	3.9	0/0/0	-	+0.20	+0.21/-2.67/-11.40	0/0	+0.21	-1.70	0/0/0	-
DINOv2-G	-1.78	3.9	0/0/0	-	+1.30	+0.24/-3.89/-11.58	0/0	+1.23	-0.93	0/0/0	-
CLIP-B	-1.95	1.3	0/0/0	-	-1.21	-1.67/-3.81/-11.38	0/0	-0.72	-1.29	0/0/0	-
CLIP-L	-0.83	1.5	0/0/3	-	-2.18	-2.43/-5.76/-13.51	0/1	-1.72	-0.68	0/0/0	-
SigLIP-B	-0.90	1.5	0/0/0	-	-1.55	-2.09/-5.04/-11.03	0/0	-1.07	-1.12	0/0/0	-

Table 10: (continued)

(c) Implanted hub alignment on the real ImageNet clouds: from the decoupled cloud of the decoupling control, each cluster's principal axis rotated toward its hub direction by a fraction s of the angle (hubs and within-cluster spectra unchanged); detection rate at $z \leq -2$ over 5 seeds per backbone and pooled.

model	$s=0$	$s=0.25$	$s=0.5$	$s=0.75$	$s=1$
ViT-T	0.2	0.8	0.8	1.0	1.0
ViT-S	0.0	0.4	1.0	1.0	1.0
ViT-B	0.0	0.8	1.0	1.0	1.0
ViT-L	0.0	0.6	1.0	1.0	1.0
DINO-B	0.0	0.0	0.0	0.0	0.8
DINOv2-S	0.0	0.0	0.0	0.0	0.0
DINOv2-B	0.0	0.0	0.0	0.0	0.0
DINOv2-L	0.0	0.0	0.0	0.2	0.0
DINOv2-G	0.0	0.0	0.0	0.2	0.4
CLIP-B	0.0	0.0	1.0	1.0	1.0
CLIP-L	0.0	0.2	0.2	0.4	0.4
SigLIP-B	0.0	0.0	1.0	0.8	0.6
pooled (12 backbones $\times$ 5 seeds)	0.02	0.23	0.50	0.55	0.60

Table 10: (continued)

(d) A deep hierarchy at the real noise level: synthetic clouds with ViT-L's real ImageNet spectrum and a three-level implanted hierarchy (nested 2/6/30 cuts of the frame) at ViT-L's real within/between ratio, a flat two-level control (30 iid hubs, same spectrum and ratio), 5 seeds each, and the WordNet Poincaré embeddings of Nickel and Kiela trained on the transitive closure of the tree over the 1000 ImageNet leaves, read with Euclidean distances on the ball coordinates.

cloud	seed	d	excess (r)	depth z	decoupled z (cert.)
three-level hierarchy, ViT-L spectrum	0	1024	-0.0081 (199)	-1.80	-4.82 ± 0.73 (1.0)
three-level hierarchy, ViT-L spectrum	1	1024	-0.0095 (199)	-2.45	-4.44 ± 0.58 (1.0)
three-level hierarchy, ViT-L spectrum	2	1024	-0.0090 (199)	-2.38	-4.19 ± 0.60 (1.0)
three-level hierarchy, ViT-L spectrum	3	1024	-0.0099 (199)	-2.08	-4.48 ± 0.67 (1.0)
three-level hierarchy, ViT-L spectrum	4	1024	-0.0090 (199)	-2.20	-4.28 ± 0.54 (1.0)
flat control, ViT-L spectrum	0	1024	-0.0198 (200)	-0.08	-1.61 ± 0.27 (0.1)
flat control, ViT-L spectrum	1	1024	-0.0202 (200)	-0.32	-1.61 ± 0.31 (0.1)
flat control, ViT-L spectrum	2	1024	-0.0196 (200)	-0.20	-1.76 ± 0.37 (0.2)
flat control, ViT-L spectrum	3	1024	-0.0197 (200)	-0.69	-1.74 ± 0.33 (0.2)
flat control, ViT-L spectrum	4	1024	-0.0192 (200)	-0.27	-1.68 ± 0.39 (0.2)
WordNet Poincaré, $d=10$	0	10	+0.0830 (0)	+2.36	$+0.32 \pm 0.58$ (0.0)
WordNet Poincaré, $d=50$	0	50	+0.0354 (0)	+1.53	-2.22 ± 0.65 (0.6)

Table 10: (continued)

(e) Power per backbone for a deep hierarchy at the backbone's own noise level: the three-level synthetic hierarchy of (d) built with each backbone's real ImageNet spectrum and within/between ratio; hierarchy test at $K=30$ (10 star seeds) on the intact cloud and the decoupling control (10 seeds) on each.

model	within/between	seeds	power	mean z	decoupled power	decoupled mean z
ViT-T	1.55	5	0.00	-0.48	0.60	-2.17
ViT-S	1.93	5	0.00	-1.04	0.90	-2.77
ViT-B	1.96	5	0.00	-1.74	1.00	-4.21
ViT-L	1.88	5	0.80	-2.18	1.00	-4.44
DINO-B	2.12	5	0.00	-1.21	0.28	-1.64
DINOv2-S	3.00	20	0.00	-0.98	0.01	-0.90
DINOv2-B	3.71	20	0.45	-1.93	0.83	-2.44
DINOv2-L	3.88	20	0.95	-2.66	1.00	-3.19
DINOv2-G	3.92	20	1.00	-2.88	1.00	-3.54
CLIP-B	1.33	5	0.20	-1.69	1.00	-5.00
CLIP-L	1.49	5	1.00	-2.83	1.00	-4.98
SigLIP-B	1.54	5	1.00	-2.49	1.00	-4.86

Table 10: (continued)

(f) False alarms of the balanced frame: the flat control of (d), ViT-L's spectrum and ratio, 5 seeds, read on the balanced thirty-superclass frame intact and decoupled (10 seeds each), with the flat hubs assigned by the balanced frame itself or by the WordNet-30 frame of record.

flat control	intact fired	mean z	decoupled fired	mean z	range
matched hubs (balanced frame)	0/5	-0.46	0/50	-0.65	+0.26 to -1.52
mismatched hubs (WordNet-30 frame)	5/5	-3.13	38/50	-2.33	-1.56 to -3.75

Table 10: (continued)

B.9 IS THE ISLAND REAL? THE TREE MAP UNDER SIX CONFIGURATIONS

Threshold sensitivity of the selection criterion of §5.4: the selected configuration is stable for degeneracy bounds in $[0.45, 0.50]$ on ImageNet and $[0.40, 0.50]$ on CIFAR-100; tighter bounds select cosine-complete on ImageNet (gap 0.13) and cosine-Ward on CIFAR-100 (gap 0.56), and in every case the label-free gap is moderate and configuration-sensitive, never a categorical island. The criterion must be applied per dataset: the configuration it selects on DBpedia (Euclidean, average linkage, where nothing is degenerate) is exactly the degenerate naive configuration on ImageNet, so a criterion transferred across datasets would reintroduce the island.

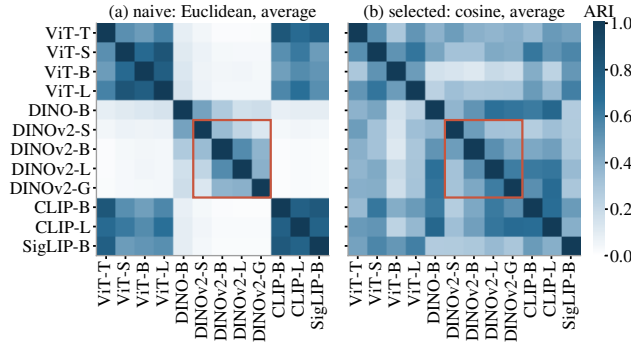


Figure 7: **The island is an artifact of the naive cut.** Pairwise ARI between dendrogram cuts under (a) the naive configuration, Euclidean distances with average linkage, where the DINOv2 block agrees with the rest at 0.03, and (b) the selected one, cosine with average linkage, where it agrees at 0.38.

(a) Selection diagnostics and mean pairwise ARI at the reference cut.

dataset	configuration	max frac.	CPCC	mean pairwise ARI at the reference cut		
				B/L/G vs block	S, DINO-B vs block	within
ImageNet	euclid-average (deg.)	0.97	0.761	0.03	0.08	0.64
ImageNet	euclid-complete (deg.)	0.83	0.606	0.03	0.17	0.44
ImageNet	euclid-ward	0.48	0.464	0.15	0.38	0.53
ImageNet	cosine-average (selected)	0.40	0.691	0.38	0.35	0.48
ImageNet	cosine-complete	0.32	0.539	0.13	0.17	0.50
ImageNet	cosine-ward	0.37	0.464	0.17	0.39	0.55
CIFAR-100	euclid-average (deg.)	0.80	0.798	0.00	0.41	0.57
CIFAR-100	euclid-complete (deg.)	0.66	0.722	0.17	0.40	0.54
CIFAR-100	euclid-ward	0.33	0.534	0.37	0.65	0.63
CIFAR-100	cosine-average (deg.)	0.53	0.777	0.30	0.54	0.58
CIFAR-100	cosine-complete (selected)	0.39	0.698	0.39	0.51	0.63
CIFAR-100	cosine-ward	0.14	0.584	0.56	0.62	0.61

Table 11: **The categorical island is a chaining artifact of the naive cut; a moderate gap remains under every admissible configuration and closes only in triplet agreement under cosine-average.** (a) Six configurations (metric $\times$ linkage) per dataset with the two quantities that drive the selection criterion of §5.4: the degeneracy diagnostic (largest-cluster fraction at the reference cut, worst model) and the mean cophenetic fidelity (CPCC) of each model’s dendrogram to its own distance matrix; configurations with max-cluster fraction > 0.5 are excluded and, among the rest, the highest CPCC selects the marked row; then the mean pairwise ARI between the cuts at 30 (ImageNet) or 20 (CIFAR-100) clusters. (b) Dendrograms rebuilt from the census cache: ARI at the cut, Pearson correlation between the two trees’ cophenetic distance vectors, and agreement on which pair merges first over 10^4 random class triplets. Under the naive ImageNet configuration, cross-family ARI between the supervised ViTs and the contrastive VLMs reaches 0.82 while every DINOv2-vs-block pair stays at or below 0.06, and the worst model’s largest cluster holds 97% of the classes. Under the selected configuration the DINOv2 family agrees with the block at 0.38 against 0.48 within the block. Under the other 3 admissible ImageNet configurations the DINOv2-vs-block agreement is 0.13–0.17 against 0.50–0.55 within the block; over the admissible configurations and the three measures the median fraction of within-block agreement missing is 0.34, about a third. (c) Two resamples of the same model agree on 0.90 of the triplets on average, 0.85 for the lowest backbone: the ceiling against which the cross-model 0.77 and the within-block 0.76 are read.

(b) Three measures of DINOv2-B/L/G against the supervised+contrastive block, each written as vs block / within block.

dataset	configuration	ARI at the cut	cophenetic corr.	triplet agreement
ImageNet	euclid-average (deg.)	0.03 / 0.64	0.36 / 0.80	0.47 / 0.74
ImageNet	euclid-complete (deg.)	0.03 / 0.44	0.20 / 0.66	0.41 / 0.65
ImageNet	euclid-ward	0.16 / 0.53	0.62 / 0.84	0.67 / 0.85
ImageNet	cosine-average (selected)	0.39 / 0.48	0.48 / 0.78	0.77 / 0.76
ImageNet	cosine-complete	0.13 / 0.49	0.18 / 0.72	0.36 / 0.71
ImageNet	cosine-ward	0.17 / 0.55	0.59 / 0.85	0.69 / 0.86
CIFAR-100	euclid-average (deg.)	0.00 / 0.57	0.15 / 0.78	0.41 / 0.74
CIFAR-100	euclid-complete (deg.)	0.17 / 0.54	0.23 / 0.72	0.41 / 0.66
CIFAR-100	euclid-ward	0.37 / 0.63	0.63 / 0.77	0.70 / 0.79
CIFAR-100	cosine-average (deg.)	0.30 / 0.58	0.42 / 0.72	0.51 / 0.64
CIFAR-100	cosine-complete (selected)	0.39 / 0.63	0.46 / 0.64	0.50 / 0.61
CIFAR-100	cosine-ward	0.56 / 0.61	0.74 / 0.73	0.76 / 0.76

Table 11: (continued)

(c) Within-model ceiling under the selected ImageNet configuration: the three measures of (b) between pairs of the 30 bootstrap centroid sets of each backbone (mean $\pm$ s.d. over the 435 pairs) and the lowest triplet agreement over pairs.

model	ARI at the cut	cophenetic corr.	triplet agreement	lowest triplet
ViT-T	0.81 ± 0.03	0.94 ± 0.02	0.93 ± 0.04	0.85
ViT-S	0.80 ± 0.11	0.94 ± 0.01	0.88 ± 0.08	0.70
ViT-B	0.94 ± 0.01	0.94 ± 0.01	0.86 ± 0.04	0.75
ViT-L	0.84 ± 0.06	0.96 ± 0.01	0.96 ± 0.02	0.87
DINO-B	0.78 ± 0.10	0.94 ± 0.01	0.96 ± 0.01	0.95
DINOv2-S	0.66 ± 0.05	0.91 ± 0.01	0.86 ± 0.05	0.69
DINOv2-B	0.64 ± 0.09	0.93 ± 0.01	0.86 ± 0.03	0.77
DINOv2-L	0.78 ± 0.05	0.94 ± 0.01	0.85 ± 0.04	0.71
DINOv2-G	0.76 ± 0.05	0.95 ± 0.01	0.88 ± 0.02	0.84
CLIP-B	0.87 ± 0.04	0.94 ± 0.01	0.93 ± 0.02	0.88
CLIP-L	0.82 ± 0.06	0.97 ± 0.01	0.95 ± 0.01	0.92
SigLIP-B	0.79 ± 0.07	0.95 ± 0.02	0.91 ± 0.03	0.84
mean over backbones	0.79 ± 0.08	0.94 ± 0.01	0.90 ± 0.04	0.85

Table 11: (continued)

B.10 WHOSE TAXONOMY? WORDNET ALIGNMENT, SUPERCLASS RECOVERY, DBPEDIA AND HIERARCAPS

On HierarCaps (Alper & Averbuch-Elor, 2024) (1000 four-level caption chains), text encoders show the same reading with no class set: the projection radius orders levels within a chain, sibling triplets are resolved at 0.83–0.97 (chance 0.5), and the leaf excess repeats the text census (Table 12).

(a) Alignment with WordNet on ImageNet.

model	Spearman ρ , inter-centroid vs WordNet distance				raw $\hat{\delta}$ of WordNet / random groups
	ρ_{WN}	shuffle	Gaussian	single image	
ViT-T	+0.52	+0.02	+0.01	–	0.146 / 0.159
ViT-S	+0.50	+0.00	–0.00	–	0.110 / 0.130
ViT-B	+0.49	+0.03	+0.01	–	0.092 / 0.120
ViT-L	+0.53	+0.01	+0.03	+0.25	0.104 / 0.117
DINO-B	+0.36	–0.01	–0.01	–	0.084 / 0.117
DINOv2-S	+0.22	+0.01	+0.01	–	0.077 / 0.100
DINOv2-B	+0.18	–0.01	+0.01	–	0.068 / 0.069
DINOv2-L	+0.18	–0.00	–0.03	+0.08	0.060 / 0.058
DINOv2-G	+0.20	–0.03	–0.01	+0.03	0.054 / 0.056
CLIP-B	+0.57	+0.00	+0.01	–	0.126 / 0.174
CLIP-L	+0.59	+0.01	+0.03	+0.26	0.117 / 0.162
SigLIP-B	+0.58	–0.01	+0.01	–	0.118 / 0.158
DeiT-B (IN-1k)	+0.08	+0.01	–	–	–
ViT-B (augreg, IN-1k)	+0.52	+0.01	–	–	–

Table 12: **Agreement with the human taxonomy follows supervision and recipe, the superclasses are recovered by every family, and the recovery is not WordNet circularity.** (a) ρ_{WN} : Spearman correlation between inter-centroid and WordNet tree distances on ImageNet, with the label-shuffle and Gaussian-cloud controls; single image: the same correlation on single-image distances (one image per class); WordNet / random groups: raw $\hat{\delta}$ of WordNet-coherent against random class groupings. (b) ARI of the 20-cluster cut against the true CIFAR-100 superclasses per configuration (degenerate configurations marked; the criterion-selected one is cosine-complete). Paired within each of the 3 admissible configurations, the label-supervised ViT-B recovers more than each DINOv2 model in 11 of 12 comparisons; DINOv2 itself recovers the superclasses at ARI 0.19–0.52, and the best cut over models and configurations reaches 0.61 against zero for permuted labels. Circularity controls: WordNet-coherent groupings read a lower raw δ than random ones in 11 of 12 models, and single-image distances still correlate with WordNet at up to +0.26. The two leaf-label ViT-B/16 of Table 9: the augreg recipe agrees with WordNet like the IN-21k ViTs (ρ_{WN} +0.52) while DeiT-B barely agrees with it (+0.08) yet recovers the CIFAR-100 superclasses at ARI up to 0.56, so agreement with WordNet is recipe-dependent even among leaf-supervised ViTs.

(b) Recovery of the CIFAR-100 superclasses: ARI of the 20-cluster cut per configuration ([†]: degenerate on CIFAR-100, euclid-average being the naive configuration; *: criterion-selected).

model	euclid-avg [†]	euclid-ward	cosine-avg [†]	cosine-complete*	cosine-ward
ViT-T	0.31	0.38	0.29	0.33	0.33
ViT-S	0.35	0.56	0.37	0.47	0.56
ViT-B	0.39	0.61	0.43	0.45	0.57
ViT-L	0.38	0.55	0.49	0.48	0.60
DINO-B	0.22	0.48	0.28	0.38	0.54
DINOv2-S	0.12	0.51	0.32	0.33	0.49
DINOv2-B	0.02	0.47	0.35	0.42	0.51
DINOv2-L	0.01	0.32	0.25	0.50	0.48
DINOv2-G	0.00	0.19	0.08	0.20	0.52
CLIP-B	0.18	0.48	0.18	0.37	0.48
CLIP-L	0.35	0.52	0.32	0.41	0.54
SigLIP-B	0.22	0.49	0.22	0.37	0.52
DeiT-B (IN-1k)	0.22	0.56	0.23	0.38	0.54
ViT-B (augreg, IN-1k)	0.30	0.44	0.40	0.42	0.48

Table 12: (continued)

(c) DBpedia Classes (219 leaf classes, 3 levels), a text hierarchy independent of WordNet; gains in pp.

model	supremum, Haar		census of record					
	$\hat{\delta}_{\max}$	excess	$\hat{\delta}_{99.9}$	excess	$r(p)$	trip.	NC H-R	FS H-R
BGE-base	0.122	-0.032	0.082	-0.021	200 (0.005)	0.89	-0.59	+0.05±0.05
E5-base	0.127	-0.028	0.086	-0.017	200 (0.005)	0.89	-0.43	+0.04±0.03
GTE-base	0.132	-0.024	0.088	-0.017	200 (0.005)	0.89	-0.82	+0.08±0.06

Table 12: (continued) (c) Excess under the original supremum reading (3 Gaussian replicates) and under the census of record (Haar null, 99.9th-percentile statistic, 200 replicates; r = replicates above the real value, left-tail p), sibling-triplet agreement, and the metric gains, marginal as the diagnostic predicts. Tree map on the same classes: no configuration degenerates (largest cluster $\leq 10\%$), the embedders recover the true 70-way level-2 partition at ARI 0.21–0.29 in all 6 configurations and agree with each other at cross-model ARI 0.65–0.82; the configuration the criterion selects there (Euclidean, average linkage) is the one that degenerates on ImageNet, which is why the criterion is applied per dataset. (d) Radial ordering of the four levels under the paper’s projection (chance for strict monotonicity = $1/24 \approx 4.2\%$; shuffle controls ≈ 0), sibling triplet agreement (cosine; chance 0.5), and leaf $\hat{\delta}$ with spectrum-null excess.

(d) HierarCaps (1000 four-level caption chains): a multi-level text hierarchy with no class set.

model	$\rho(\text{level, radius})$	% monotone	trip. L1	trip. L2	$\hat{\delta}$ leaves	excess
BGE-base	+0.80±0.31	49.6	0.88	0.97	0.130	+0.003
E5-base	+0.87±0.19	53.9	0.90	0.97	0.139	+0.010
GTE-base	+0.47±0.55	24.0	0.90	0.97	0.135	+0.006
CLIP-B	+0.64±0.43	27.2	0.86	0.95	0.123	-0.011
CLIP-L	+0.51±0.49	18.7	0.86	0.96	0.117	-0.010
SigLIP-B	+0.32±0.55	9.8	0.83	0.91	0.138	+0.014

Table 12: (continued)

B.11 IS LOCAL STRUCTURE SHARED? PERMUTATION-CALIBRATED AGREEMENT

measure	raw	null mean	$\tau_{0.05}$	calibrated	null-centered	pairs $p < 0.05$
mutual-kNN ($k=10$), Euclidean	0.472	0.010	0.015	0.464	0.462	66/66
mutual-kNN, Poincaré	0.425	0.010	0.019	0.414	0.415	66/66
linear CKA, Euclidean	0.633	0.108	0.111	0.577	0.524	66/66
linear CKA, Poincaré	0.640	0.108	0.111	0.586	0.532	66/66

Table 13: **Cross-model agreement survives permutation calibration.** Means over the 66 model pairs of the 12 vision backbones on the 1000 ImageNet class centroids (census cache). Calibration of Gröger et al. (2026) with $K=200$ permutations of the class correspondence and $\alpha=0.05$, scalar (no layer search): $\tau_{0.05}$ is the $\lceil 0.95(K+1) \rceil$ -th order statistic of the observed score and its nulls (eq. 9), $p = (1 + \#\{\text{null} \geq \text{obs}\})/(K+1)$ (eq. 10), calibrated = $\max\{(\text{obs} - \tau_{0.05})/(1 - \tau_{0.05}), 0\}$ (eq. 12); “null-centered” is the earlier raw-minus-null-mean variant, kept for continuity. The mKNN null mean equals the analytic chance level $k/(n-1) = 10/999$.

B.12 WHAT FOLLOWS FOR PRACTICE? THE COROLLARY

One parameter-free projection and two tasks serve the corollary. The parameter-free map $\phi(x) = \tanh(\frac{1}{2}\|s(x - \mu)\|) \cdot \frac{s(x - \mu)}{\|s(x - \mu)\|}$ centers features, scales the 95th-percentile norm to radius $1/\sqrt{2}$, and preserves directions; prototypes in the ball are means of projected points re-mapped through the origin. Tasks: nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes); metric arms: Euclidean (R), Poincaré (H), the radial map with Euclidean distances (RT), cosine (COS), and the L2-normalized stack. Changing the readout metric on the intact configuration, which is what the rule selects among, is distinct from deforming the configuration itself.

Both readings predict the gain within datasets. Within each hierarchical dataset the Poincaré advantage tracks the record excess: the nearest-centroid correlation is -0.73 on ImageNet with a confidence interval that excludes zero on all four datasets, the few-shot correlation excludes zero on CIFAR-100, CIFAR-10 and DTD but not on ImageNet, and the hierarchy-test z predicts no gain anywhere (Table 14). The raw reading correlates as strongly (-0.84 on ImageNet for nearest-centroid) and survives partialing out dimension, baseline accuracy, family and spectral covariates; without DINOv2 it holds on ImageNet and CIFAR-100 but not on CIFAR-10/DTD. Pooled cross-dataset correlations are family-driven and not used as evidence; on flat-label datasets no metric helps. Gains carry paired confidence intervals and a McNemar test.

The training objective picks the metric. The radial map with Euclidean distances changes nothing, so the gain is the metric itself, and the split follows the objective: cosine suffices for DINOv2, Poincaré $\approx$ cosine for supervised ViTs, and Poincaré beats cosine for CLIP by up to $+1.3\text{pp}$ even after L2 normalization (Table 14). The rule is read off the families it is scored on: it describes them rather than validating a policy.

A diagnostic, not a policy. Khrulkov et al. (2020) estimate $c \approx 0.33$ from the raw reading of their datasets; the same rule on structureless Gaussian clouds gives the curvatures of Section 6. Cosine is a safe default, gating on the reading is the weakest policy tested, and gains are modest and limited to prototype-based tasks (sample-level tasks never benefit, Table 8). On DBpedia, a three-level text hierarchy, the diagnostic predicts correctly on both sides: genuine excess, no metric gain.

(a) Task accuracies and zero-cost metric gains per cell (10 backbones with cached per-dataset features $\times$ 6 datasets).

model	data	NC acc. (%)		FS acc. (%)		FS advantage over Euclidean (pp)			McNemar
		Eucl.	Poinc.	Eucl.	Poinc.	cosine	radial map	Poinc. _N - cos	NC H vs R
ViT-T	IN	68.1	67.2	97.5	97.6	-0.01	+0.01	+0.08	10^{-22} (-)
	C100	29.3	28.2	62.4	63.1	+1.01	+0.45	+0.55	10^{-5} (-)
	C10	87.5	87.4	87.6	88.1	+0.66	+0.19	+0.29	0.19 (-)
	DTD	63.7	62.5	83.3	83.9	-0.23	+0.32	+0.69	0.03 (-)
	FMNIST	78.5	78.4	81.6	81.9	+0.21	+0.10	+0.31	0.38 (-)
	MNIST	75.9	74.0	74.3	74.6	+0.64	+0.52	+0.20	10^{-13} (-)
ViT-S	IN	77.7	77.5	98.6	98.7	+0.17	-0.02	-0.07	0.09 (-)
	C100	74.9	74.6	92.2	92.4	+0.59	+0.16	-0.00	0.10 (-)
	C10	93.4	93.2	92.0	92.5	+0.56	+0.27	+0.19	0.02 (-)
	DTD	69.3	68.5	85.4	86.0	-0.03	+0.25	+0.73	0.10 (-)
	FMNIST	80.7	80.4	83.7	83.5	+0.28	+0.08	-0.19	0.08 (-)
	MNIST	84.8	84.1	80.5	81.4	+0.34	+0.65	+0.84	10^{-4} (-)
ViT-B	IN	81.6	81.7	98.9	99.0	+0.20	-0.04	-0.05	0.21 (+)
	C100	77.3	77.0	92.5	92.7	+0.56	+0.12	-0.05	0.07 (-)
	C10	94.5	94.0	92.7	93.0	+0.49	+0.18	+0.15	10^{-5} (-)
	DTD	70.8	70.8	85.4	86.2	+0.28	+0.19	+0.69	1.00 (-)
	FMNIST	81.3	81.6	83.4	83.3	+0.03	+0.09	-0.04	0.08 (+)
	MNIST	81.6	80.5	76.8	77.6	+0.19	+0.51	+0.66	10^{-7} (-)
ViT-L	IN	86.1	86.2	99.4	99.4	+0.11	-0.03	-0.02	0.58 (+)
	C100	85.3	85.2	94.8	95.5	+1.17	+0.31	+0.12	0.59 (-)
	C10	97.5	97.5	95.0	95.8	+1.25	+0.35	+0.59	0.76 (+)
	DTD	71.8	71.5	85.6	86.5	+0.13	+0.16	+1.02	0.69 (-)
	FMNIST	83.7	83.5	84.4	84.7	+0.00	+0.17	+0.23	0.12 (-)
	MNIST	84.2	83.6	80.8	81.6	+0.08	+0.42	+0.67	0.00 (-)
DINOv2-S	IN	74.7	74.4	97.4	97.9	+0.68	+0.03	-0.22	0.00 (-)
	C100	73.4	74.4	94.3	95.1	+0.80	+0.15	+0.06	10^{-6} (+)
	C10	95.0	95.2	92.9	93.5	-0.36	+0.13	+1.06	0.14 (+)
	DTD	73.4	73.4	89.8	90.5	+0.30	+0.04	+0.36	1.00 (+)
	FMNIST	82.2	82.2	85.4	85.5	+0.02	+0.07	+0.07	1.00 (-)
	MNIST	83.4	81.9	80.2	80.8	-0.07	+0.45	+0.67	10^{-11} (-)
DINOv2-B	IN	78.9	78.9	97.0	97.8	+1.40	+0.04	-0.63	0.57 (+)
	C100	87.4	87.9	96.5	97.2	+1.11	+0.09	-0.34	10^{-4} (+)
	C10	97.1	97.4	94.7	95.4	+0.27	+0.03	+0.43	10^{-4} (+)
	DTD	74.2	74.7	89.2	90.3	+1.48	+0.01	-0.35	0.31 (+)
	FMNIST	83.2	82.8	85.9	85.9	+0.21	+0.04	-0.18	0.01 (-)
	MNIST	80.6	78.8	76.6	77.1	+0.17	+0.48	+0.47	10^{-16} (-)
DINOv2-L	IN	79.8	80.4	96.2	97.1	+2.13	+0.02	-1.24	10^{-14} (+)
	C100	85.6	87.7	96.1	97.1	+1.80	+0.09	-0.74	10^{-38} (+)
	C10	97.9	98.3	94.1	95.1	+0.92	-0.00	+0.11	10^{-8} (+)
	DTD	74.4	75.2	87.1	88.4	+2.40	+0.03	-0.98	0.11 (+)
	FMNIST	84.4	84.5	87.8	87.9	+0.29	+0.03	-0.03	0.40 (+)
	MNIST	74.1	72.7	70.5	71.1	+0.25	+0.43	+0.46	10^{-12} (-)
DINOv2-G	IN	78.4	78.3	96.2	97.1	+2.57	+0.03	-1.65	0.10 (-)
	C100	89.8	91.0	95.9	97.2	+2.44	+0.13	-1.10	10^{-16} (+)
	C10	97.8	98.2	92.3	93.8	+2.51	-0.02	-1.01	10^{-13} (+)
	DTD	72.9	73.1	87.2	88.5	+2.63	+0.01	-1.38	0.81 (+)
	FMNIST	85.9	85.8	89.0	89.2	+0.38	+0.04	-0.07	0.72 (-)
	MNIST	79.3	78.6	72.9	73.8	+0.30	+0.56	+0.72	0.00 (-)
CLIP-B	IN	61.4	59.9	96.8	96.8	-0.04	-0.00	+0.14	10^{-40} (-)
	C100	65.6	65.1	87.7	88.9	-0.03	+0.44	+1.21	0.07 (-)
	C10	87.9	88.3	89.5	90.1	-0.19	+0.20	+0.90	10^{-4} (+)
	DTD	63.7	63.2	82.2	83.1	-0.23	+0.36	+1.29	0.46 (-)
	FMNIST	75.9	75.1	78.8	78.9	+0.06	+0.17	+0.10	10^{-4} (-)
	MNIST	70.1	66.6	70.9	69.7	+0.04	+0.23	-1.17	10^{-45} (-)
CLIP-L	IN	75.0	74.5	98.4	98.6	+0.01	+0.00	+0.22	10^{-8} (-)
	C100	74.6	74.5	90.0	90.8	+0.12	+0.39	+1.23	0.66 (-)
	C10	95.9	95.9	92.4	93.0	+0.16	+0.25	+0.91	0.59 (-)
	DTD	68.6	69.0	85.2	86.2	-0.08	+0.36	+1.24	0.45 (+)
	FMNIST	76.2	75.9	76.8	77.3	+0.12	+0.33	+0.68	0.09 (-)
	MNIST	85.2	86.1	83.7	84.7	+0.08	+0.58	+1.06	10^{-7} (+)

Table 14: **Both readings predict the zero-cost gain within datasets, the hierarchy verdict predicts nothing, and cosine is the safe default.** (a) Nearest-centroid (NC) and 5-way 5-shot (FS, 1000 paired episodes) accuracy under Euclidean and Poincaré readouts, regenerated from the audited re-runs; FS advantage of cosine and of the radial map with Euclidean distances (the map without the metric changes nothing) over Euclidean, and of the Poincaré readout over cosine after L2 normalization; McNemar p for the NC test-set decisions, Poincaré vs Euclidean (order of magnitude when $p < 10^{-3}$; sign of the NC advantage in parentheses).

(b) Correlation between the raw supremum δ_{norm} and the metric gain within each dataset ($n=10$ backbones), with bootstrap CI95 and the family-identity control (demeaned: both variables centered within family; within-family r for ViT and DINOv2, sign only for CLIP).

gain	dataset	r [CI95]	demeaned	ViT	DINOv2	CLIP
NC H—R	ImageNet	−0.83 [−0.97, −0.65]	−0.73	−0.99	−0.77	(−1)
	CIFAR-100	−0.87 [−0.97, −0.78]	−0.45	−0.68	−0.33	(−1)
	CIFAR-10	−0.63 [−0.97, −0.06]	−0.31	+0.01	−0.98	(+1)
	DTD	−0.46 [−0.90, +0.22]	−0.57	−0.78	−0.76	(+1)
	pooled (cluster bootstrap)	−0.45 [−0.60, −0.36]	—	—	—	—
FS H—R	ImageNet	−0.57 [−0.91, +0.14]	−0.73	−0.04	−0.97	(−1)
	CIFAR-100	−0.67 [−0.95, −0.15]	−0.63	−0.41	−0.96	(+1)
	CIFAR-10	−0.78 [−0.93, −0.23]	−0.64	−0.22	−0.79	(−1)
	DTD	−0.71 [−0.94, −0.32]	−0.86	−0.95	−0.98	(+1)
	pooled (cluster bootstrap)	−0.18 [−0.50, +0.12]	—	—	—	—
FS best—R	ImageNet	−0.64 —	−0.71	−0.57	−0.90	(−1)
	CIFAR-100	−0.83 —	−0.82	−0.58	−0.96	(+1)
	CIFAR-10	−0.63 —	−0.55	−0.26	−0.65	(−1)
	DTD	−0.60 —	−0.70	−0.95	−0.88	(+1)

Table 14: (continued) (b) Within-dataset correlations of the raw supremum reading; bootstrap CI95 from 10k resamples of the 10 models, pooled rows by cluster bootstrap over datasets; family control: both variables centered within family before correlating, and within-family correlations (ViT, DINOv2: 4 models each; CLIP has two, so only its sign is shown). Without DINOv2 (6 backbones) the raw NC correlation holds on ImageNet and CIFAR-100 (−0.87/−0.65) but not on CIFAR-10/DTD (+0.23/−0.34); pooled cross-dataset correlations of the best-metric advantage do not survive the family control (−0.31 → −0.18 over all 60 cells, −0.33 → −0.04 on hierarchical cells) and are family-driven. On the record excess the nearest-centroid prediction survives (CI95 excluding zero) on ImageNet, CIFAR-100, CIFAR-10, DTD and the few-shot prediction on CIFAR-100, CIFAR-10, DTD; the hierarchy verdict predicts no gain in any dataset. (c) The gain per backbone and which readout collects it.

(c) Best zero-cost metric per backbone: few-shot advantage over Euclidean, mean over the four hierarchical datasets, and which metric collects it (either: within 0.15pp).

model	best—R (pp)	metric
ViT-T	+0.58	either
ViT-S	+0.49	either
ViT-B	+0.51	either
ViT-L	+0.87	either
DINOv2-S	+0.68	H
DINOv2-B	+1.17	cos
DINOv2-L	+1.84	cos
DINOv2-G	+2.54	cos
CLIP-B	+0.67	H
CLIP-L	+0.65	H

Table 14: (continued)

(d) The same correlations on the raw $\hat{\delta}_{99.9}$, the record excess and the hierarchy-test z (Fisher- z CI95; dm: family-demeaned r).

gain	dataset	raw $\hat{\delta}_{99.9}$ r [CI95]	dm	record excess r [CI95]	dm	depth z r [CI95]
NC H-R	ImageNet	-0.84 [-0.96, -0.45]	-0.83	-0.73 [-0.93, -0.19]	-0.61	-0.28 [-0.77, +0.42]
	CIFAR-100	-0.92 [-0.98, -0.68]	-0.58	-0.79 [-0.95, -0.33]	-0.21	+0.52 [-0.16, +0.87]
	CIFAR-10	-0.63 [-0.90, +0.00]	-0.31	-0.67 [-0.91, -0.06]	-0.34	n/a
	DTD	-0.67 [-0.91, -0.07]	-0.67	-0.76 [-0.94, -0.26]	-0.60	n/a
FS H-R	ImageNet	-0.86 [-0.97, -0.50]	-0.56	-0.21 [-0.74, +0.48]	+0.35	+0.27 [-0.43, +0.77]
	CIFAR-100	-0.62 [-0.90, +0.02]	-0.51	-0.79 [-0.95, -0.31]	-0.80	-0.26 [-0.76, +0.45]
	CIFAR-10	-0.78 [-0.95, -0.31]	-0.64	-0.78 [-0.95, -0.30]	-0.66	n/a
	DTD	-0.87 [-0.97, -0.53]	-0.94	-0.98 [-1.00, -0.91]	-0.97	n/a
FS best-R	ImageNet	-0.83 [-0.96, -0.43]	-0.50	-0.08 [-0.68, +0.58]	+0.47	+0.13 [-0.54, +0.70]
	CIFAR-100	-0.77 [-0.94, -0.27]	-0.75	-0.75 [-0.94, -0.23]	-0.65	+0.22 [-0.48, +0.75]
	CIFAR-10	-0.63 [-0.90, -0.01]	-0.55	-0.62 [-0.90, +0.02]	-0.56	n/a
	DTD	-0.81 [-0.95, -0.36]	-0.78	-0.90 [-0.98, -0.63]	-0.84	n/a

Table 14: (continued) (d) The calibrated reading: the raw 99.9th-percentile statistic, the record excess and the hierarchy-test z of Table 9 as predictors. (e) Validation picking retains the oracle advantage and the zero-validation objective rule modestly beats always-cosine. (f) The gate’s value is avoiding the Poincaré projection on flat-label data; cosine is a safe NC default everywhere. (g) The headline $t = 1/\sqrt{2} \approx 0.71$ was fixed a priori and is conservative for FS.

(e) Few-shot metric-selection policies on held-out episodes (500 validation / 500 test per cell).

policy (test advantage over Euclidean, pp)	all 60 cells	hierarchical (40)
always cosine	+0.23	+0.28
objective rule (zero validation)	+0.28	+0.41
δ -gated objective rule	+0.08	+0.13
validation-picked per cell	+0.63	+0.78
test oracle (reference)	+0.64	+0.78

Table 14: (continued)

(f) Nearest-centroid metric-selection policies.

NC policy (advantage over Euclidean, pp)	all 60	flat (20)	hierarchical (40)
always cosine	+0.48	+0.24	+0.59
always Poincaré	-0.24	-0.70	-0.00
gated rule (flat→R)	+0.34	+0.00	+0.51
δ -gated rule	+0.19	+0.00	+0.29

Table 14: (continued)

(g) Sensitivity to the projection target t (36 hierarchical cells).

target t	0.25	0.35	0.41	0.45	0.50	0.58	0.71	0.95
FS adv. mean (pp)	+0.88	+0.87	+0.87	+0.86	+0.84	+0.82	+0.76	+0.56
NC adv. mean (pp)	-0.30	-0.25	-0.23	-0.21	-0.17	-0.12	-0.01	+0.15
% cells FS > 0	97	100	100	100	100	100	100	100

Table 14: (continued)

B.13 PROVENANCE

table	generated file	result file(s)
5	tab_q01_census_final	expR75_census_centered_haar.csv expR52_census_haar_p999_200.csv expR54_census_haar_sup_200.csv expR40b_p999census200.csv expR39c_census200_cache.csv expR57_census_cosine_haar_p999_200.csv expR57_text_cosine_haar_p999_200.csv expl0_local_vs_global.csv expR45_convnet_rows.csv exp20_null_ztable.csv
7	tab_q02_text	expR53_text_haar_p999_200.csv expR53_text_haar_sup_200.csv expR53_text_gauss_p999_200.csv expR48b_text_census200_bsl.csv expR57_text_cosine_haar_p999_200.csv expl8_text_nulls.csv exp4_prompt_variation.csv expl7_wordnet_prompts.csv expR49_template_nulls_bsl.csv expR47_extraction_variance.csv
8	tab_q03_sample_final	expR62_samplelevel_record.csv expR78_khrulkov_replication.csv
9	tab_q04_depth_final	expR78_khrulkov_replication_summary.csv expR56_depth_variants.csv expR69_depth_haarhubs_summary.csv expR64b_wn30bal_summary.csv expR70_inet1k_supervised.csv expR52_census_haar_p999_200.csv expR63_meru_record.csv expR65_hier_finetune.csv expR74_decoupling_summary.csv expR71_meru_radial.csv expR77_positive_control.csv expR77_positive_control_verdict.json expR82_radial_control.csv expR82_radial_control_summary.csv
10	tab_q05_power_final	expR55_depth_power.csv expR55b_depth_power_leafframe.csv expR64b_wn30.csv expR64b_wn30_summary.csv expR64b_wn30bal_summary.csv expR67_frame_choice.json expR69_depth_haarhubs_summary.csv expR80_implanted_alignment.csv expR80_decision.csv expR79_synthetic_deep_poincare.csv expR81_deep_per_backbone.csv expR81_deep_per_backbone_summary.csv expR83_flat_balanced.csv expR83_flat_balanced_summary.csv
11	tab_q06_treemap	exp22_tree_similarity_imagenet.npz exp23_treemap_controls.npz exp23_config_diagnostics.json expR58_treemap_cutfree_summary.csv expR84_tree_ceiling.csv expR84_tree_ceiling_summary.csv
12	tab_q07_wordnet_final	exp3_alignment.csv exp28_recovery_per_config.csv exp8_pl_recovery.csv expl_delta_controls.csv exp8_p6_pooling.csv expR70_inet1k_supervised.csv exp14_dbpedia.csv expR61_dbpedia_record.csv exp27_dbpedia_treemap.json expl6_hierarcaps.csv
3	tab_q08_robust_final	expR72_budget_record.csv expR72_budget_record_summary.csv expR59_imagenet_bootstrap_summary.csv expR73_transfer_bootstrap_record_summary.csv expR66_joint_sensitivity_summary.csv expR60_c_sweep_record.csv analysis4_finetuning.csv el_delta_by_layer.csv expR66c_joint_sensitivity_summary.csv
14	tab_q09_corollary	table1_regenerated.csv exp2_metric_controls.csv exp2b_normalized_stack.csv expl3_mcnemar.csv night/correlation_cis.csv exp20_null_ztable.csv expR68_corollary_excess.csv exp24_val_metric_selection.csv night/t_sweep.csv
4	tab_q10_calibration	expl_delta_controls.csv expR42_star_calibration.csv
13	tab_q13_local	exp21b_local_global_K200.csv
6	tab_q14_panel	(model panel: stated in the generator; parameter counts from the model cards)

Table 15: Provenance index: every appendix table is generated by script from the released per-cell result files listed here; no table value is hand-transcribed. The same mapping for main-text tables and figures is annotated in the source and shipped with the code release.